



模式识别与计算机视觉

Chapter 4: Frequency Domain and Sampling

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Three views of filtering



- Image filters in spatial domain
 - Filter is a mathematical operation on values of each patch
 - Smoothing, sharpening, measuring texture
- **Image filters in the frequency domain**
 - Filtering is a way to modify the frequencies of images
 - Denoising, sampling, image compression
- Templates and Image Pyramids
 - Filtering is a way to match a template to the image
 - Detection, coarse-to-fine registration

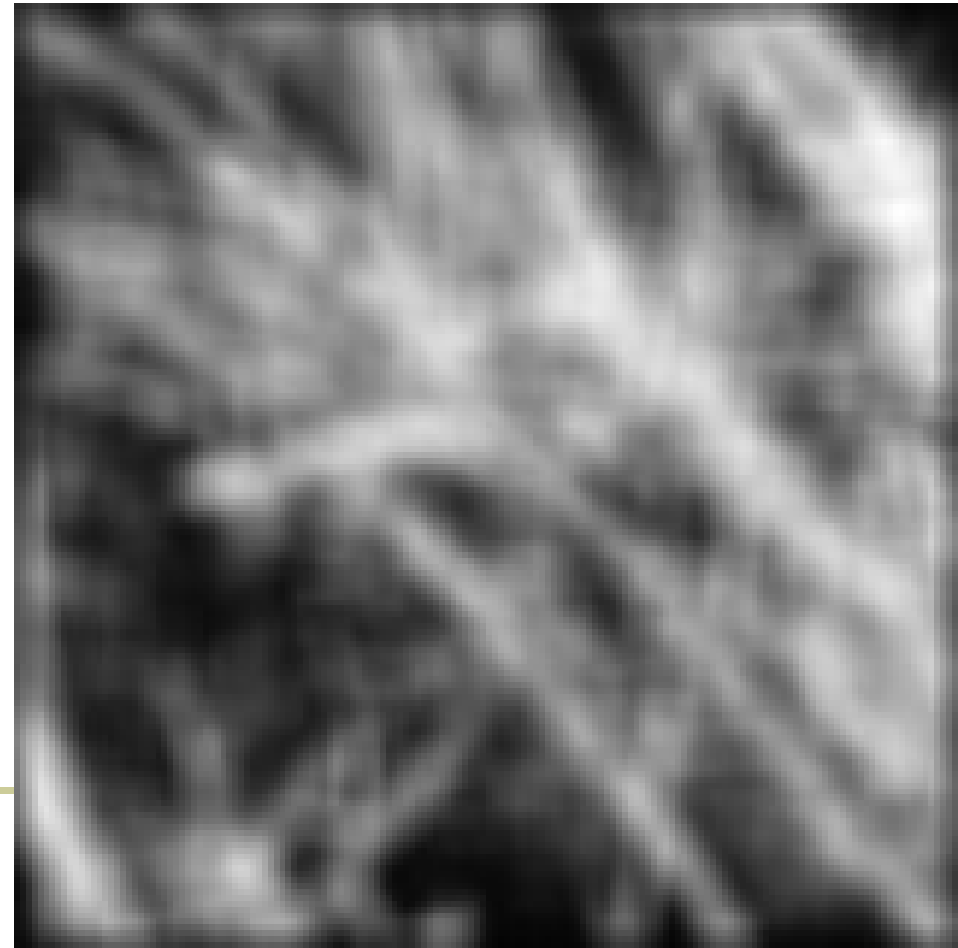
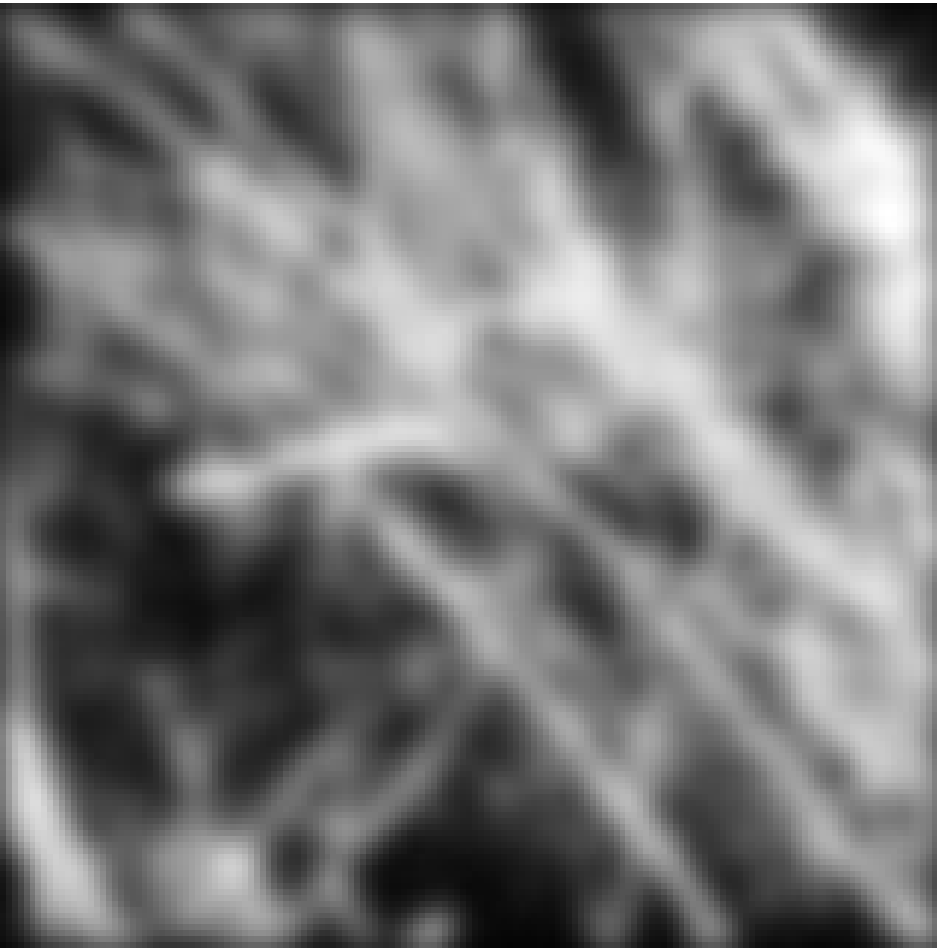


Why does the Gaussian give a nice smooth image, but the square filter give edgy artifacts?

Gaussian

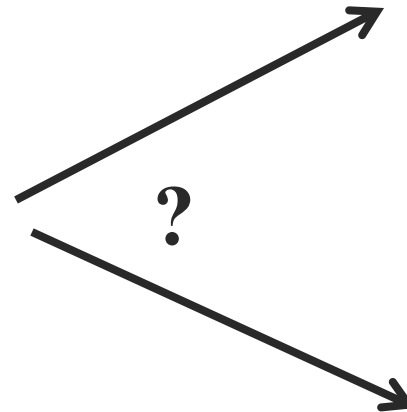


Box filter





Why do we get different, distance-dependent interpretations of hybrid images?





Why does a lower resolution image still make sense to us? What do we lose?





Today's Class



- **Frequency domain and Fourier transform**
- 1D Discrete Fourier Transform
- 2D Image Fourier Transform
- Sampling
- Hybrid Image



Jean Baptiste Joseph Fourier (1768-1830)

...the manner in which the author arrives at these equations is not exempt of difficulties and...his analysis to integrate them still leaves something to be desired on the score of generality and even rigour.

had crazy idea (1807):

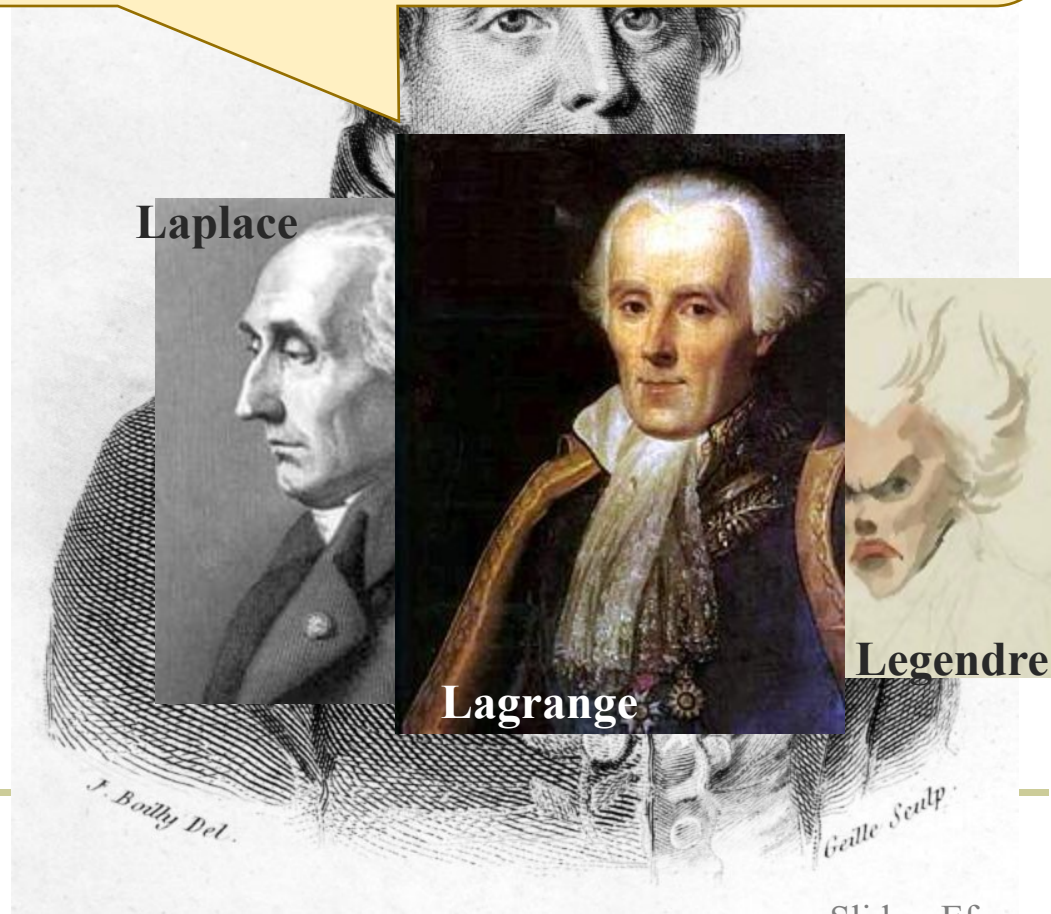
Any univariate function can be rewritten as a weighted sum of sines and cosines of different frequencies.

■ Don't believe it?

- Neither did Lagrange, Laplace, Poisson and other big wigs
- Not translated into English until 1878!

■ But it's (mostly) true!

- called Fourier Series
- there are some subtle restrictions

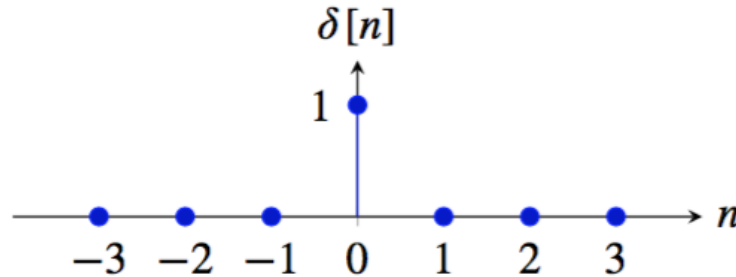




Important signals



The impulse



The result of convolving a signal $g[n]$ with the impulse signal is the same signal:

$$f[n] = \delta \circ g = \sum_k \delta[n - k] g[k] = g[n]$$

Convolving a signal f with a translated impulse $\delta[n - n_0]$ results in a translated signal:

$$f[n - n_0] = \delta[n - n_0] \circ f[n]$$



Important signals



Cosine and sine waves

$$s(t) = A \sin(w t - \theta)$$

$$s_k[n] = \sin\left(\frac{2\pi}{N} k n\right)$$

$$c_k[n] = \cos\left(\frac{2\pi}{N} k n\right)$$

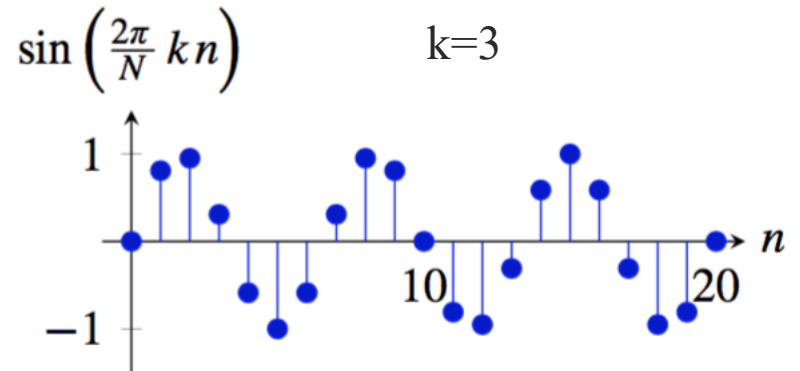
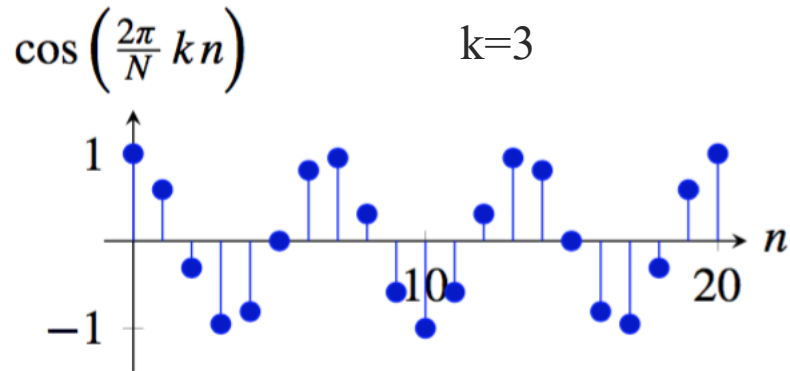
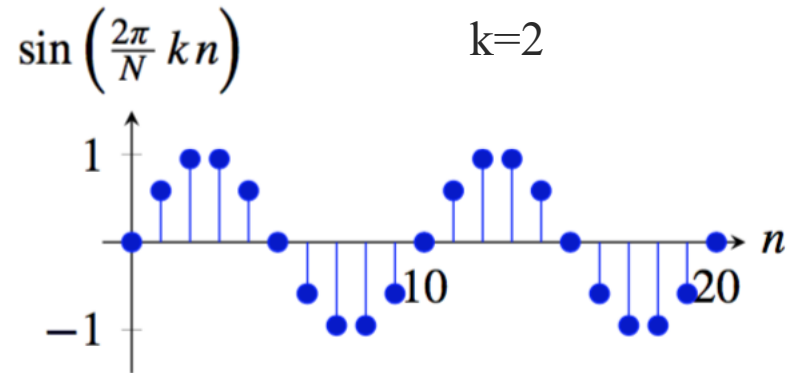
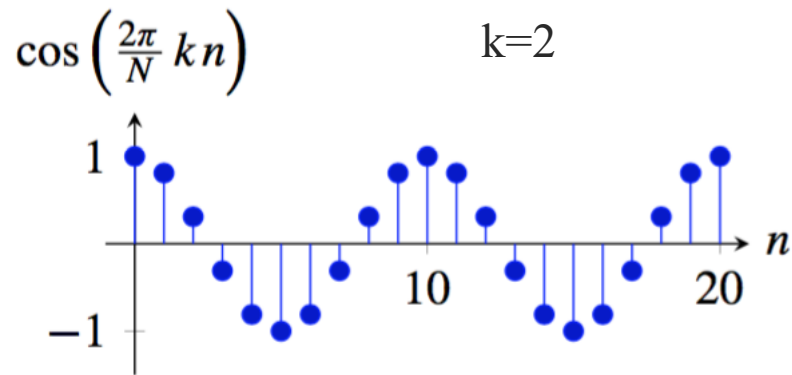
$k \in [1, N/2]$ denotes the number of wave cycles that will occur within the region of support



Important signals



Cosine and sine waves





Important signals



Complex exponential

$$s(t) = A \exp(j\omega t)$$

In discrete time (setting $A = 1$), we can write:

$$e_k[n] = \exp\left(j\frac{2\pi}{N}kn\right) = \cos\left(\frac{2\pi}{N}kn\right) + j\sin\left(\frac{2\pi}{N}kn\right)$$

And in 2D, the complex exponential wave is:

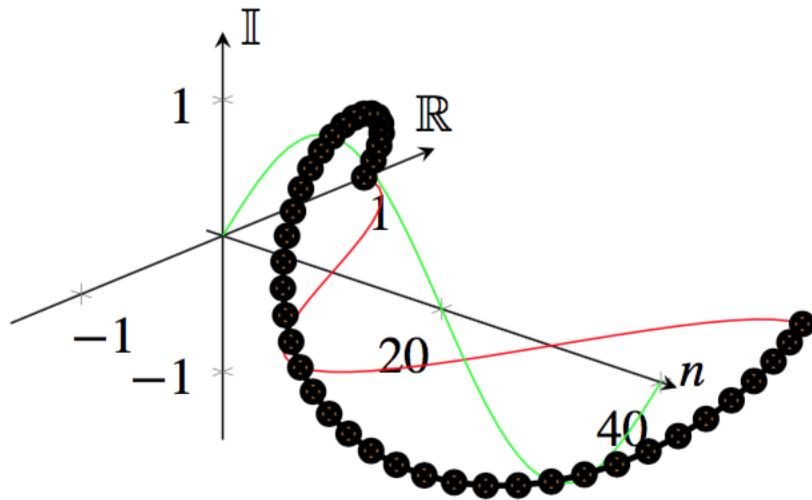
$$e_{u,v}[n,m] = \exp\left(2\pi j\left(\frac{un}{N} + \frac{vm}{M}\right)\right)$$



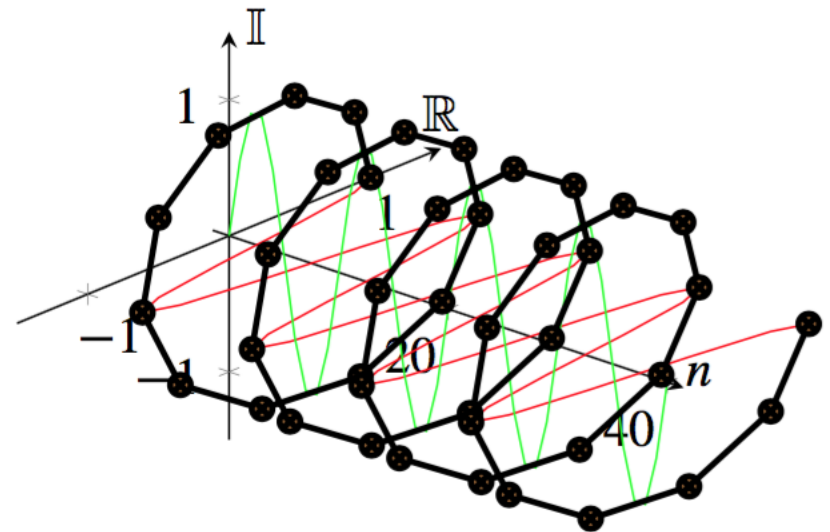
Important signals



Complex exponential



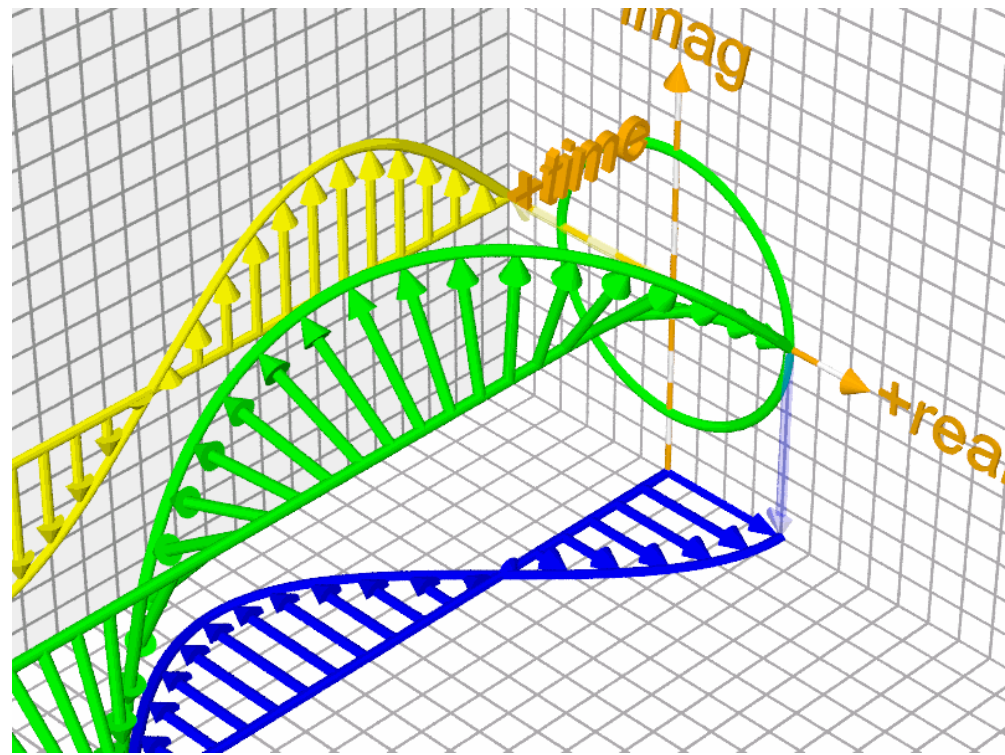
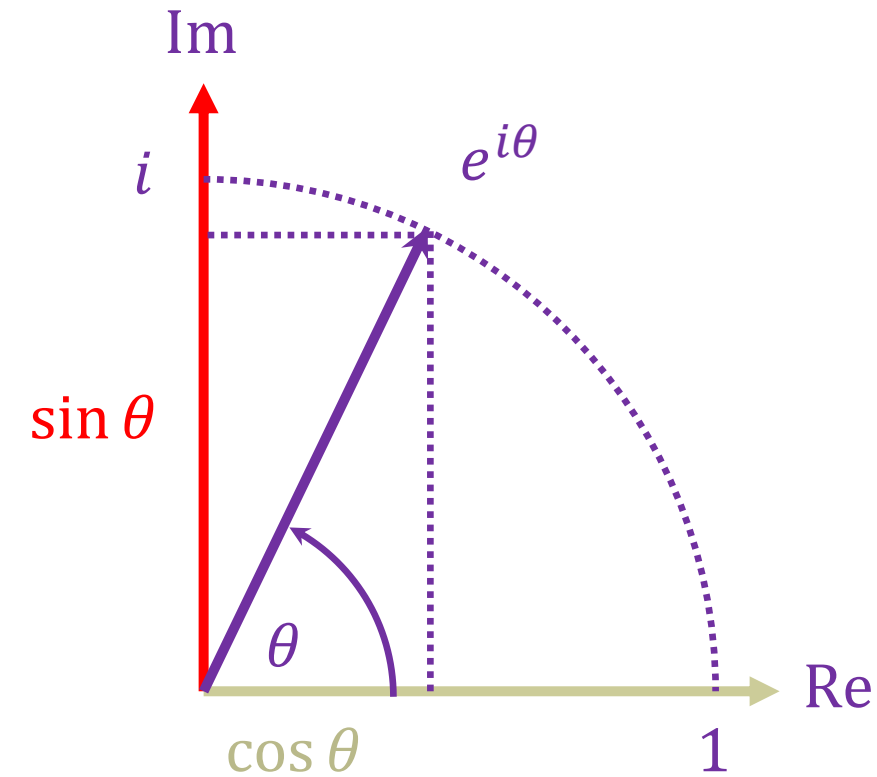
$N = 40, k = 1$



$N = 40, k = 4$



Demo





A simple idea: a sum of sines



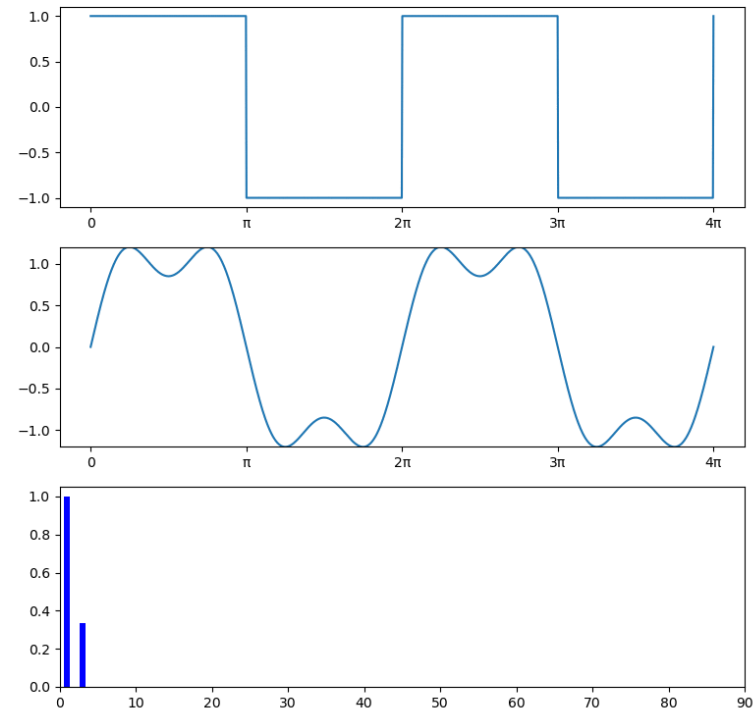
Our building block:

$$A \sin(\omega x + \phi)$$

Add enough of them to get any signal $f(x)$ you want!

Example: series for a square wave

$$\sum_{k=1,3,5,\dots}^{\infty} \frac{1}{k} \sin(kt)$$

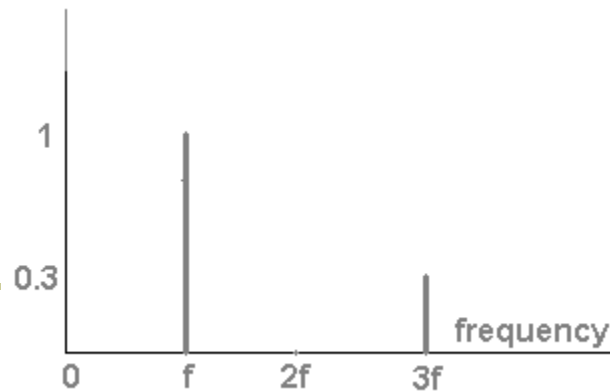
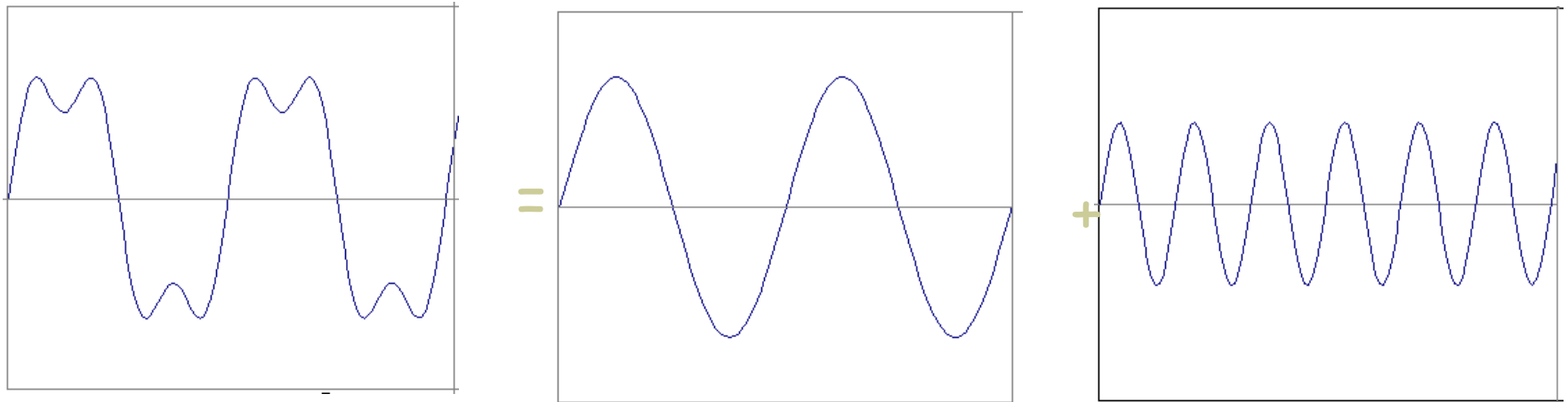




Frequency Spectra

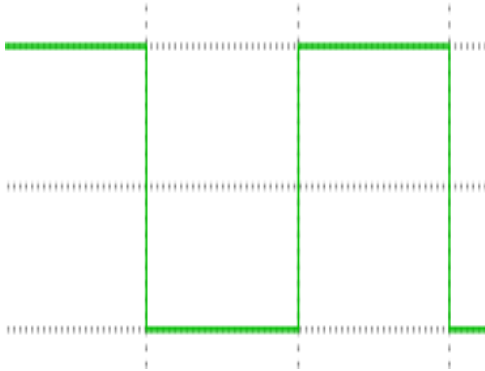


- example : $g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi(3f) t)$



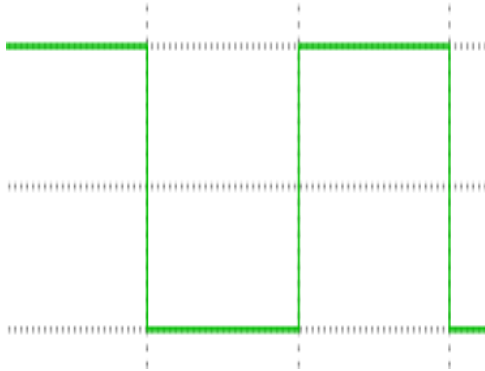


Frequency Spectra

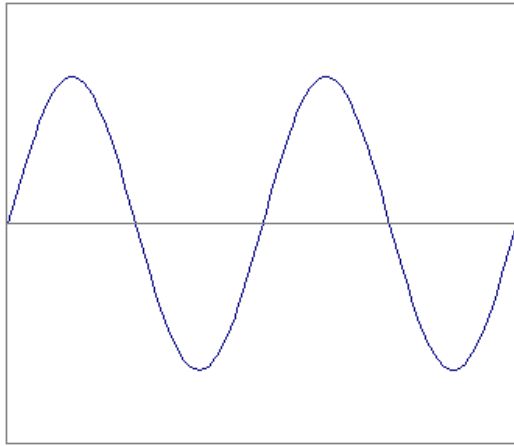




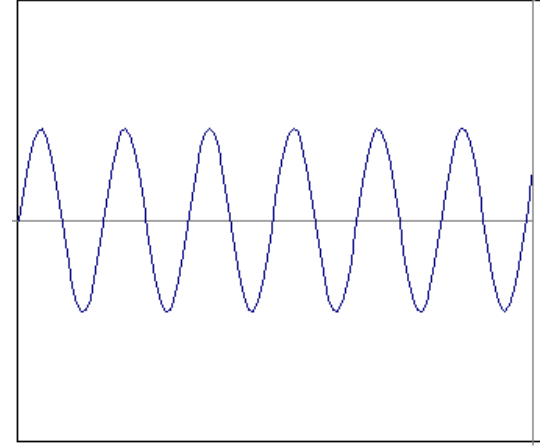
Frequency Spectra



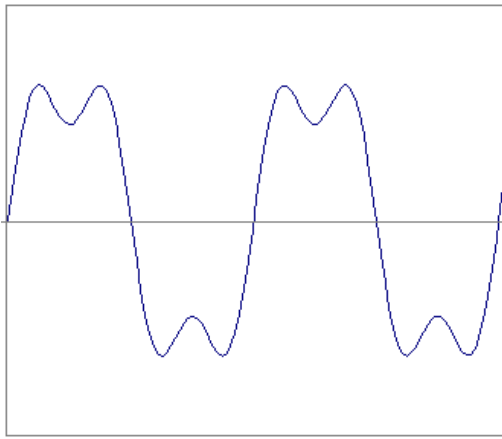
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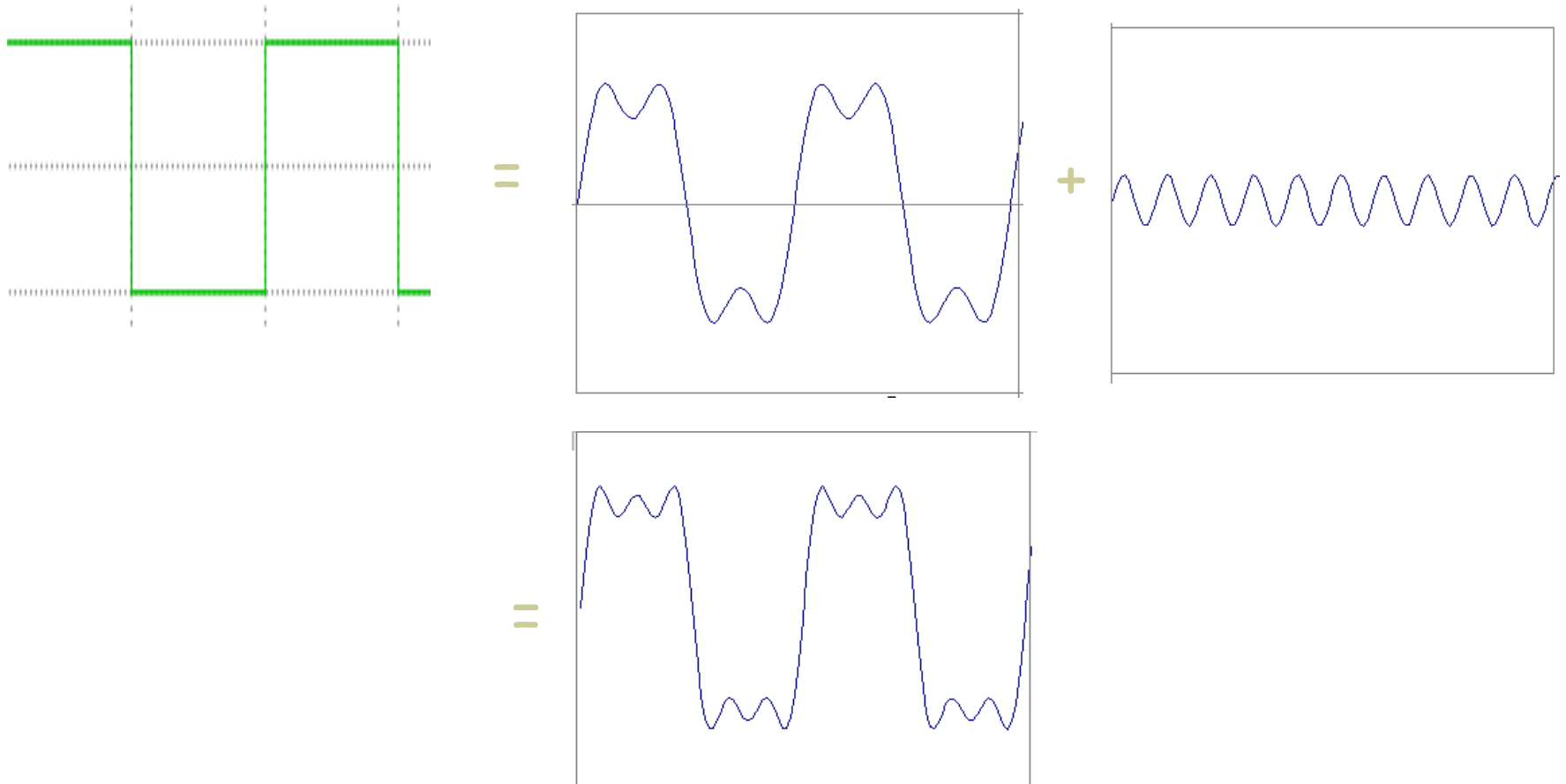


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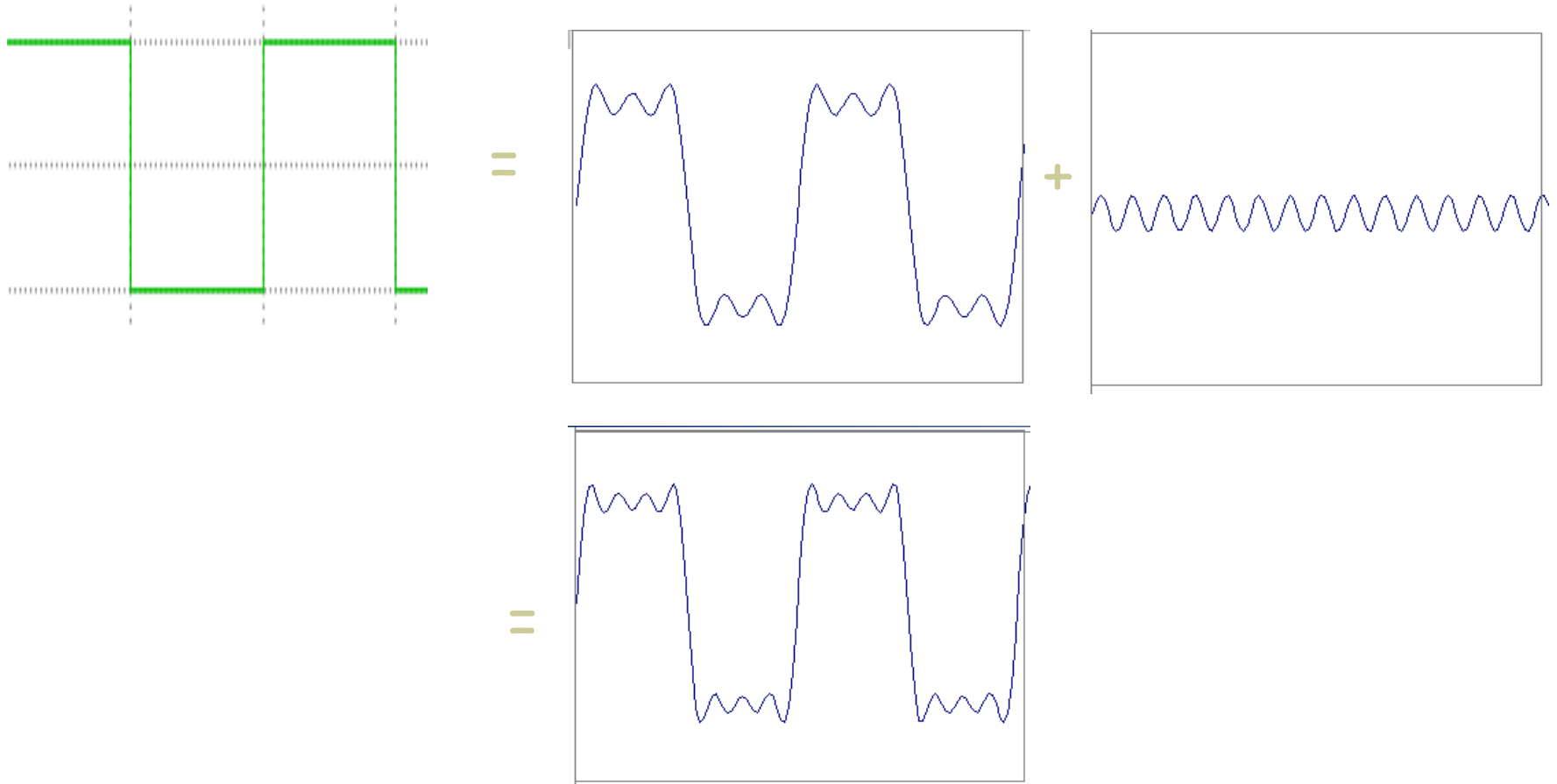


Frequency Spectra



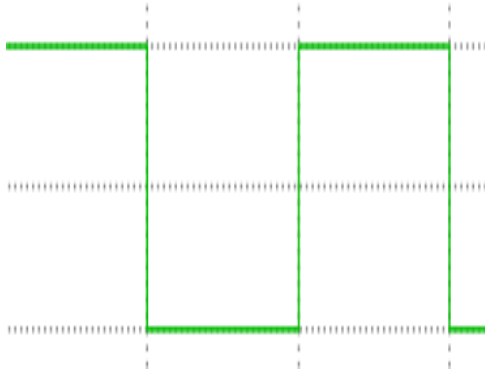


Frequency Spectra

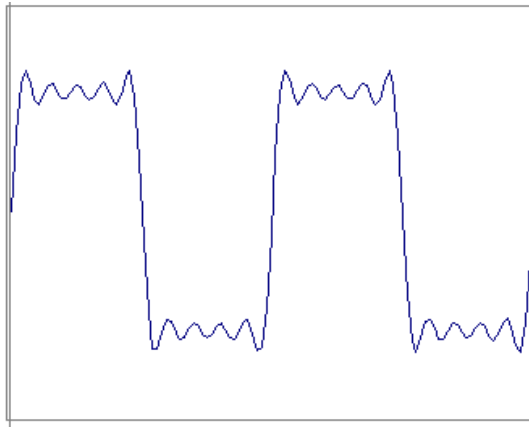




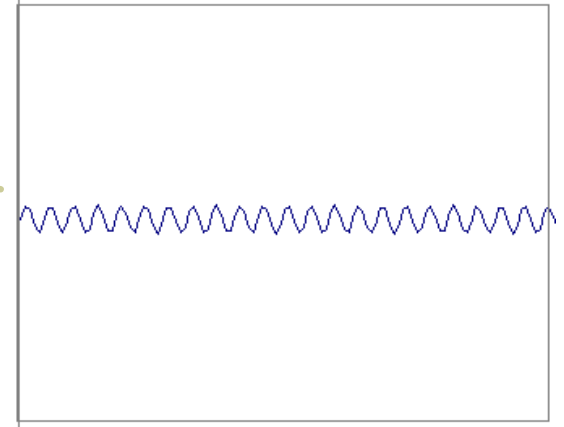
Frequency Spectra



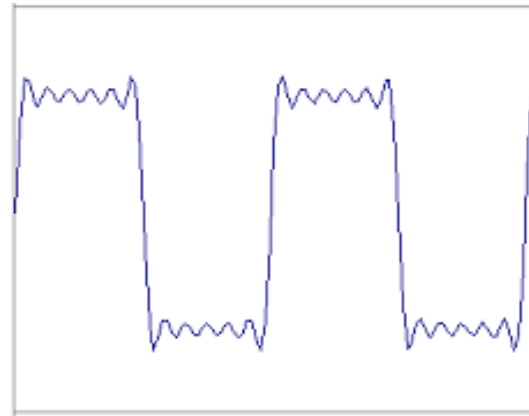
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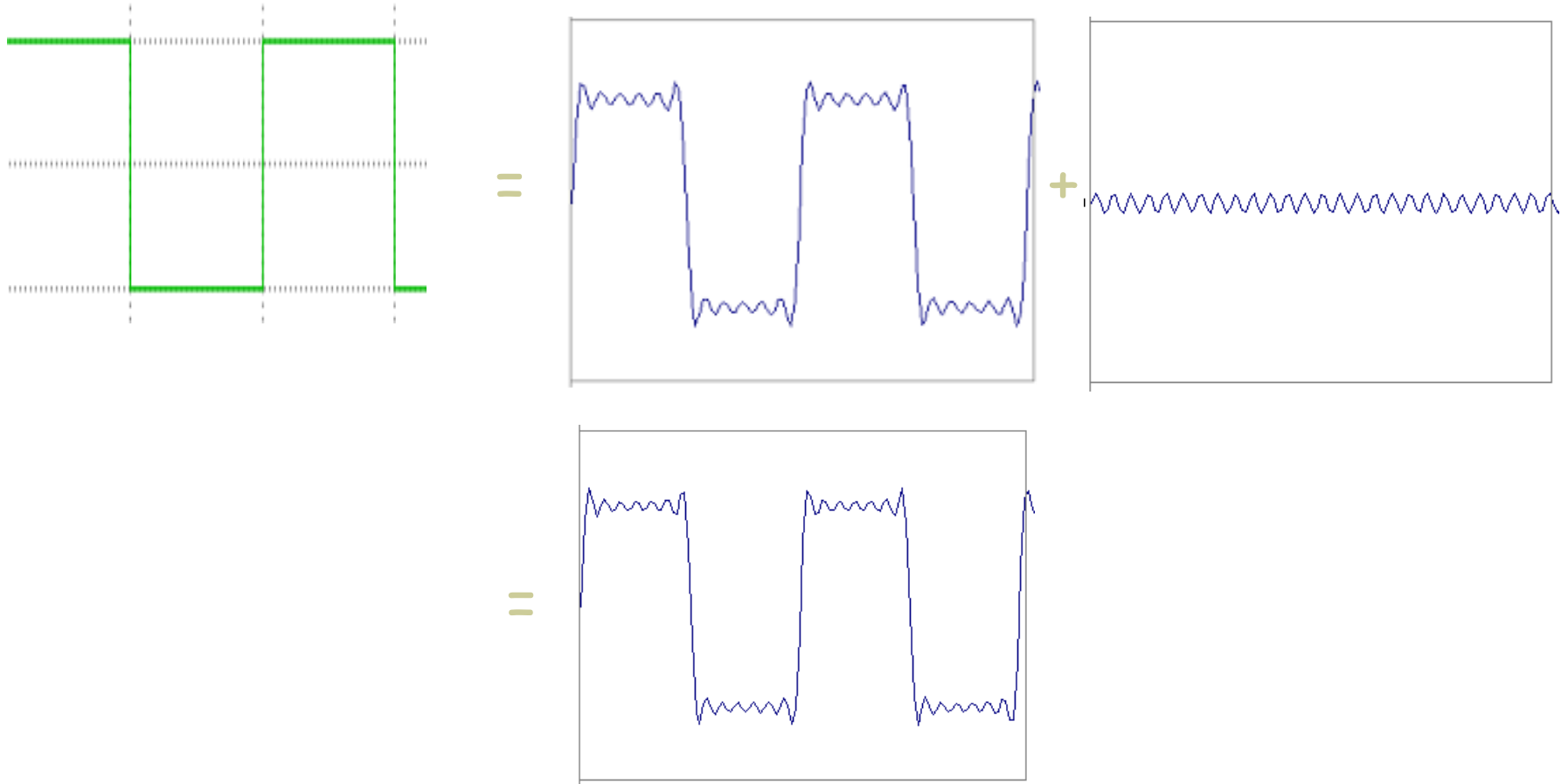


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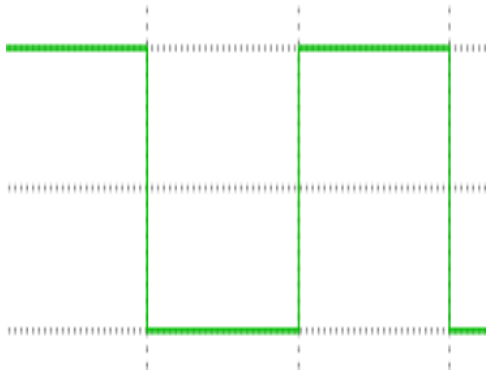


Frequency Spectra



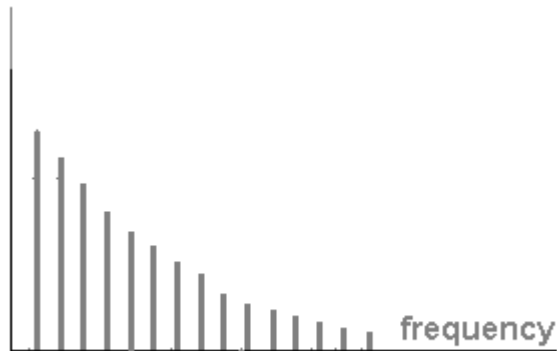


Frequency Spectra



=

$$A \sum_{k=1}^{\infty} \frac{1}{k} \sin(2\pi kt)$$





Linear Transformation



- In analyzing images, it is often useful to make a change of basis.

Transformed image $\vec{F} = U\vec{f}$ Vectorized image

U

Fourier transform, or
Wavelet transform, or
Steerable pyramid transform



Self-inverting transforms



$$\vec{F} = U\vec{f} \longleftrightarrow \vec{f} = U^{-1}\vec{F}$$

Same basis functions are used for the inverse transform

$$\begin{aligned}\vec{f} &= U^{-1}\vec{F} \\ &= U^+\vec{F}\end{aligned}$$

U transpose and complex conjugate



Today's Class



- Frequency domain and Fourier transform
- **1D Discrete Fourier Transform**
- 2D Image Fourier Transform
- Sampling
- Hybrid Image

The Discrete Fourier transform

Discrete Fourier Transform (DFT) transforms a signal $f[n]$ into $F[u]$ as:

$$F[u] = \sum_{n=0}^{N-1} f[n] \exp\left(-2\pi j \frac{un}{N}\right)$$

Discrete Fourier Transform (DFT) is a linear operator. Therefore, we can write:

The diagram illustrates the DFT as a matrix operation. On the left, a green vertical bar represents the output spectrum F . This is followed by an equals sign. In the center is a large red grid representing the DFT matrix, with the expression $\exp\left(-2\pi j \frac{un}{N}\right)$ centered within it. Above the grid, a horizontal arrow labeled n points to the right, indicating the column index. To the left of the grid, a vertical arrow labeled u points downwards, indicating the row index. The top row of the grid contains question marks, and the bottom row contains an ellipsis followed by a question mark. To the right of the grid is a blue vertical bar representing the input signal f . Below the grid, the text "NxN array" is written.

Lets visualize the transform coefficients

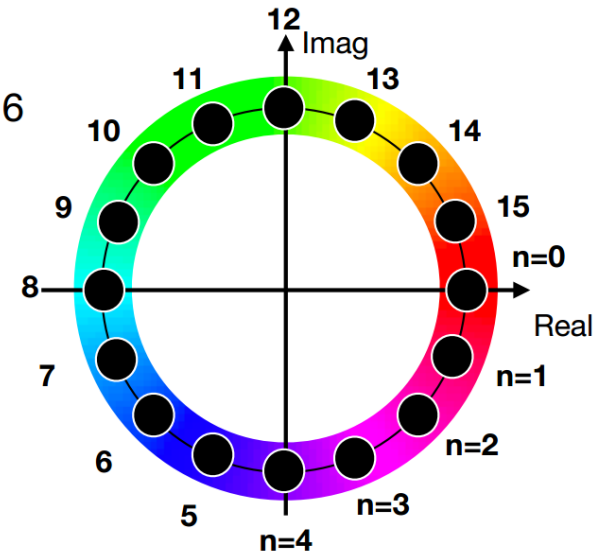
Visualizing the Fourier transform

$$F[u] = \sum_{n=0}^{N-1} f[n] \exp\left(-2\pi j \frac{un}{N}\right)$$

$$\exp(\alpha j) = \cos(\alpha) + j \sin(\alpha)$$

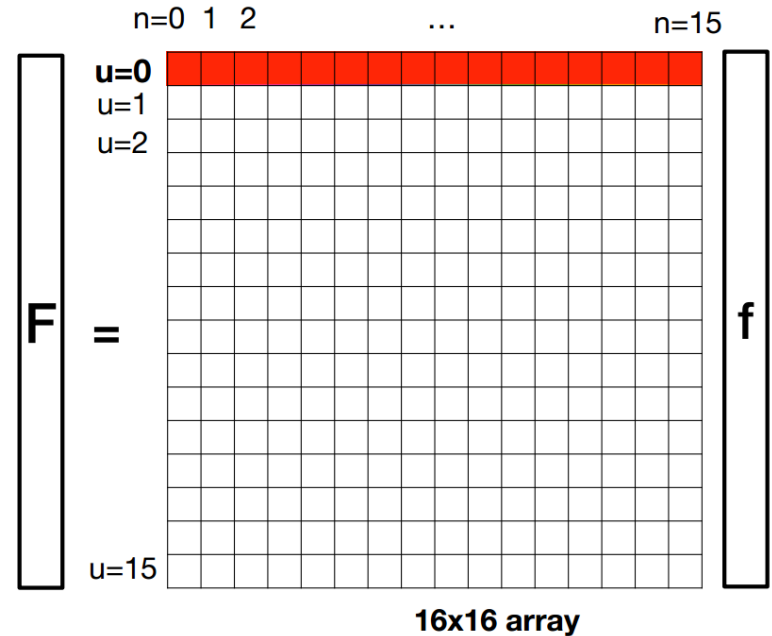
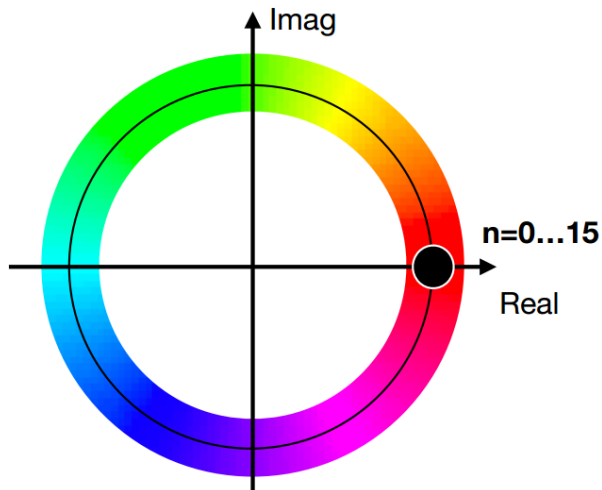
$$\cos\left(2\pi \frac{un}{N}\right) - j \sin\left(2\pi \frac{un}{N}\right)$$

For:
 $u=1$
 $N=16$



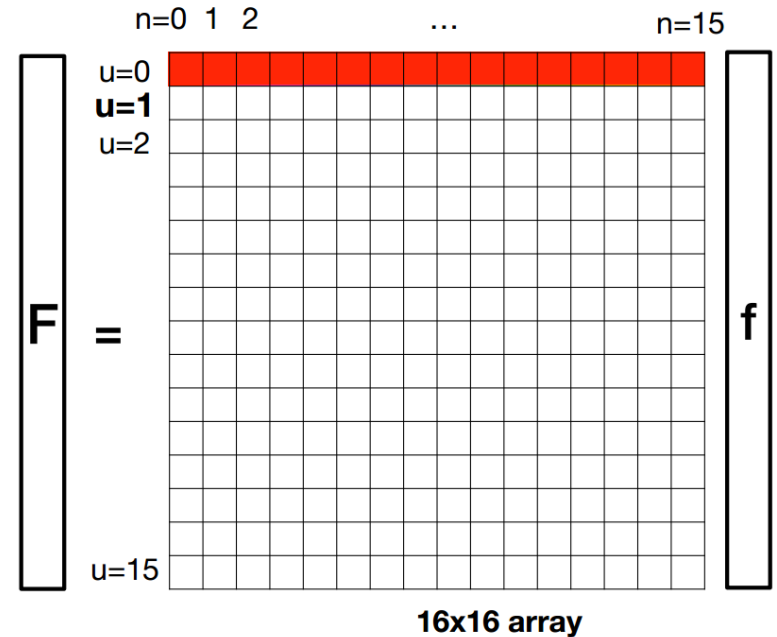
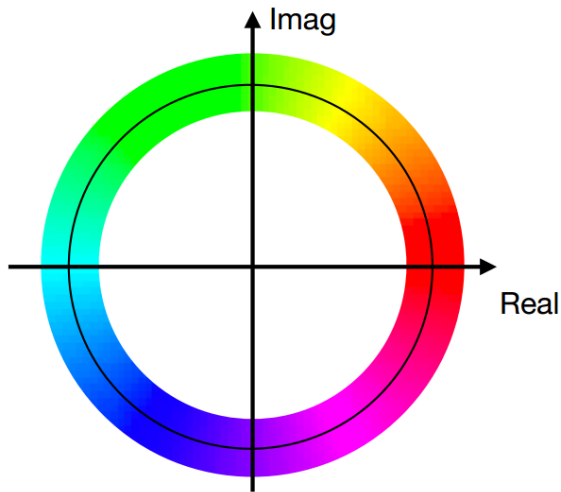
Visualizing the transform coefficients

$$\exp\left(-2\pi j \frac{un}{N}\right) \quad \text{For } N=16$$



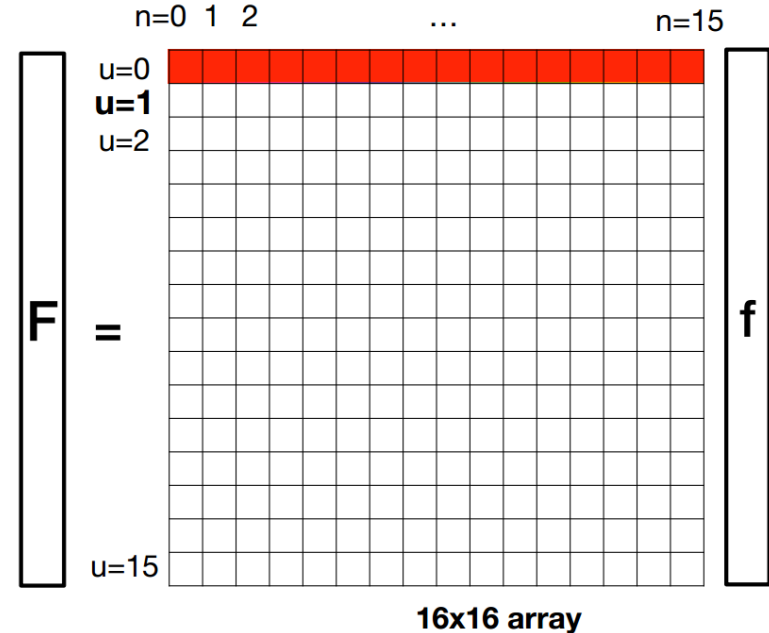
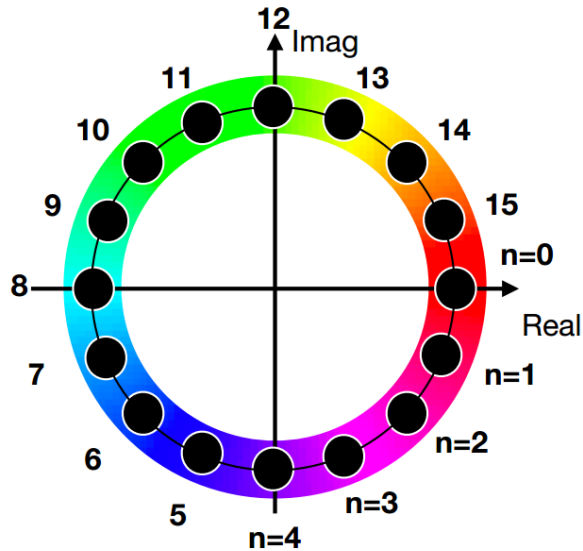
Visualizing the transform coefficients

$$\exp\left(-2\pi j \frac{un}{N}\right) \quad \text{For } N=16$$



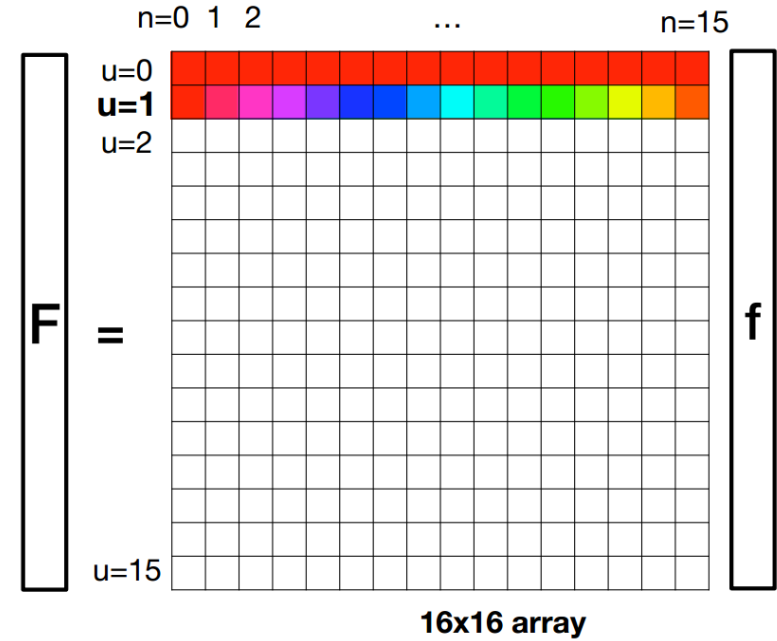
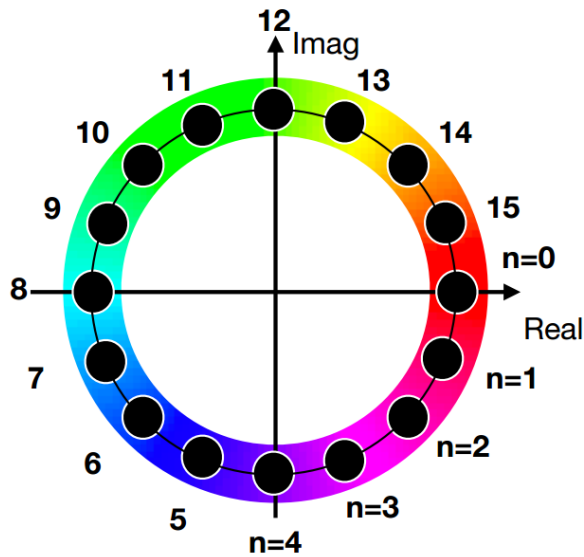
Visualizing the transform coefficients

$$\exp\left(-2\pi j \frac{un}{N}\right) \quad \text{For } N=16$$



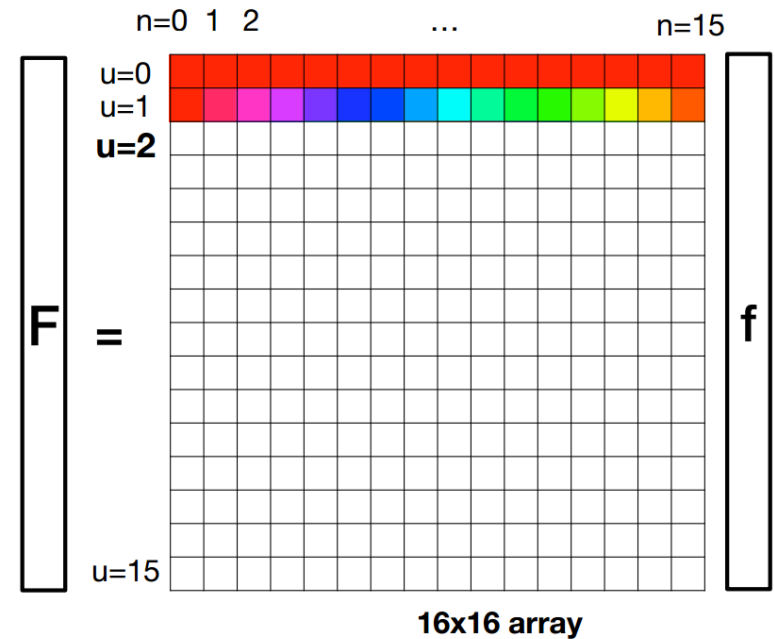
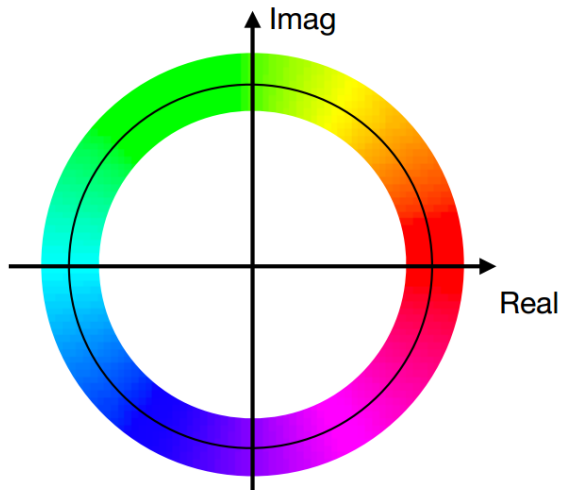
Visualizing the transform coefficients

$$\exp\left(-2\pi j \frac{un}{N}\right) \quad \text{For } N=16$$



Visualizing the transform coefficients

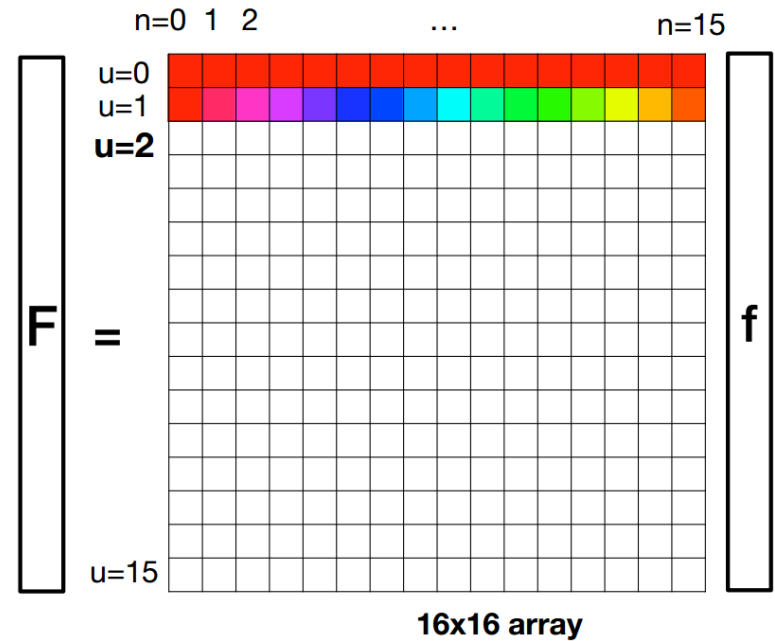
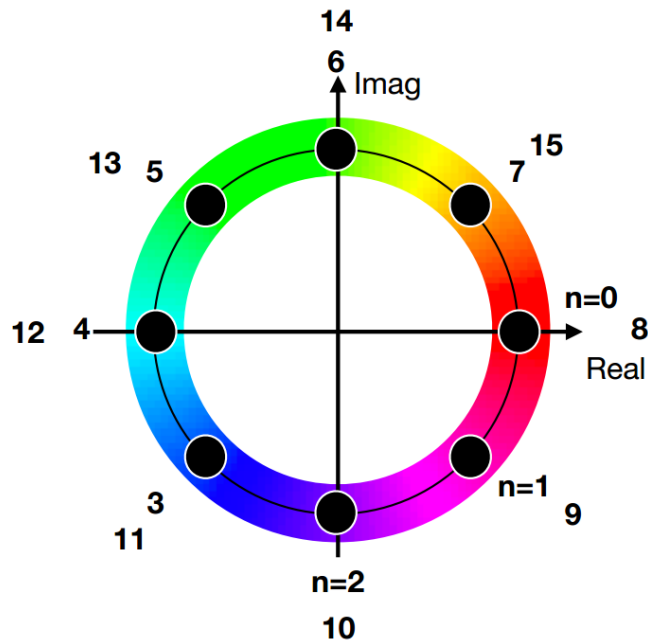
$$\exp\left(-2\pi j \frac{un}{N}\right) \quad \text{For } N=16$$



Visualizing the transform coefficients

$$\exp\left(-2\pi j \frac{un}{N}\right)$$

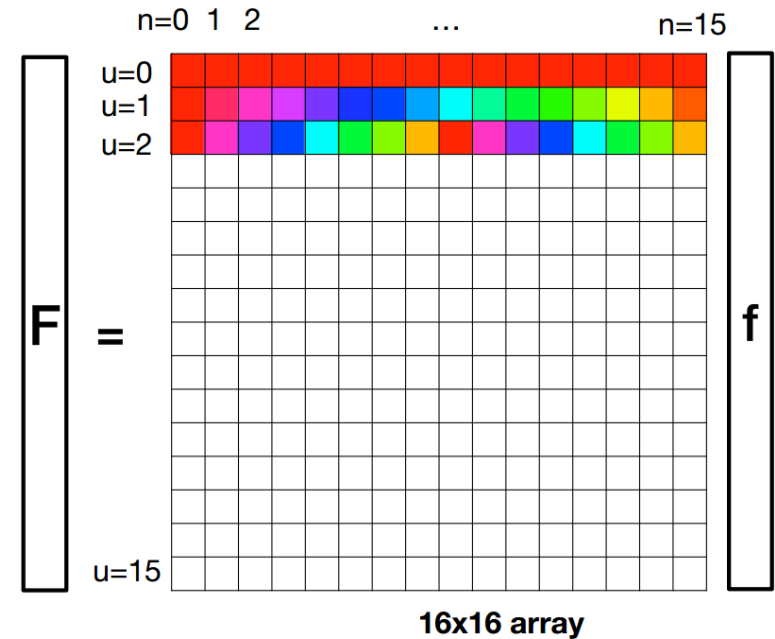
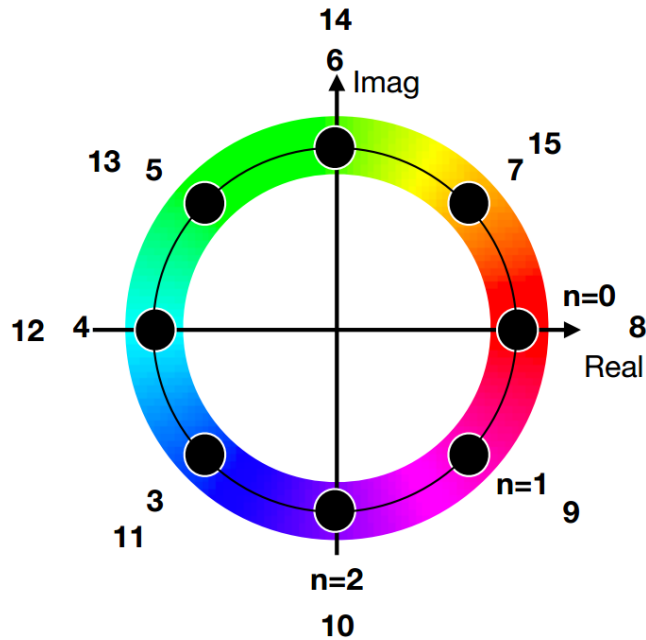
For N=16



Visualizing the transform coefficients

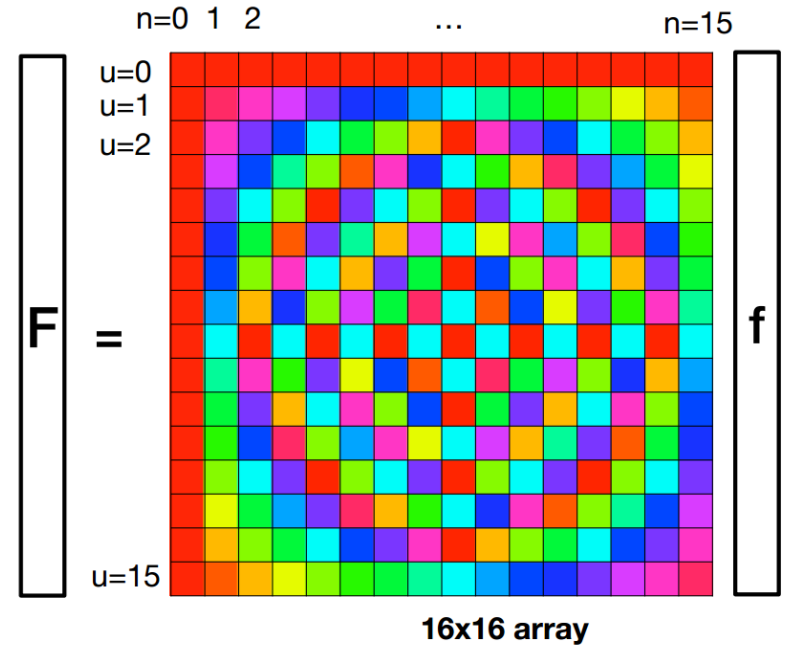
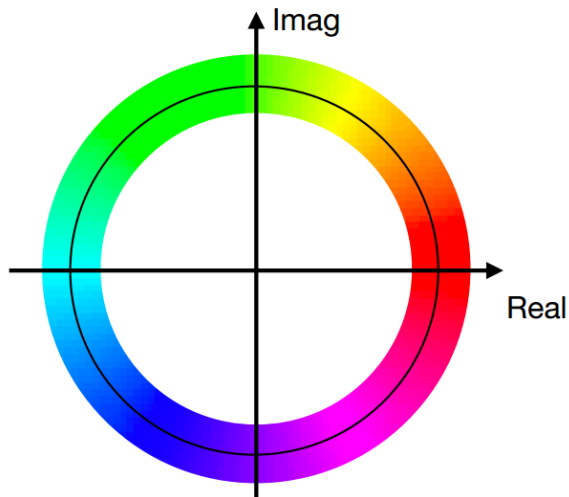
$$\exp\left(-2\pi j \frac{un}{N}\right)$$

For N=16



Visualizing the transform coefficients

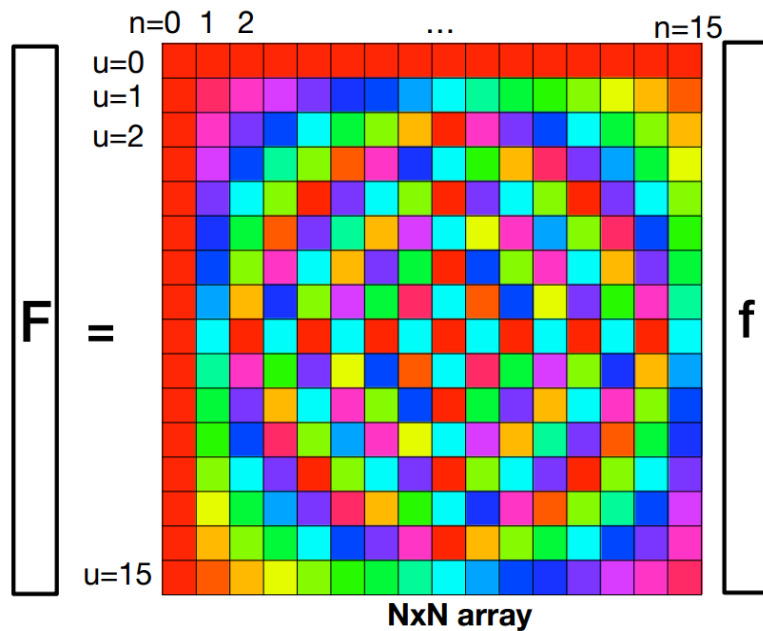
$$\exp\left(-2\pi j \frac{un}{N}\right) \quad \text{For } N=16$$



The inverse of the Discrete Fourier transform

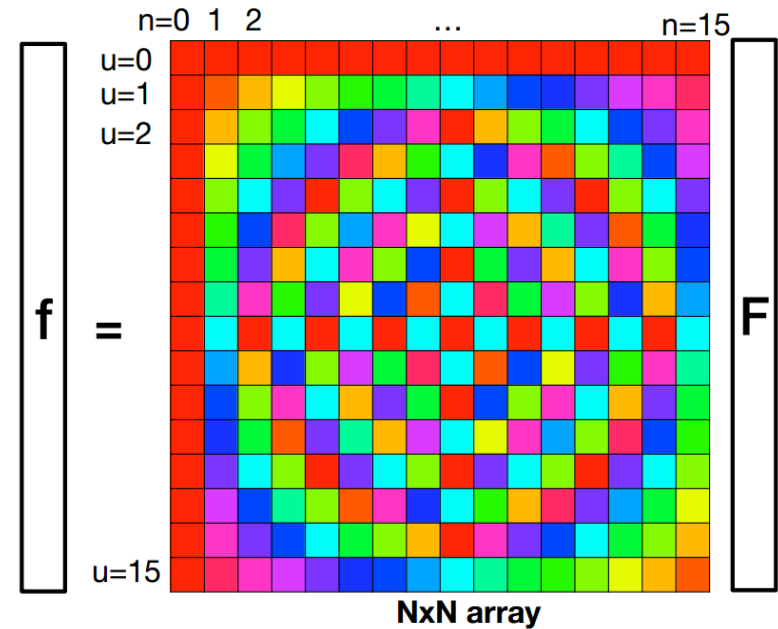
Discrete Fourier Transform (DFT):

$$F[u] = \sum_{n=0}^{N-1} f[n] \exp\left(-2\pi j \frac{un}{N}\right)$$



Its inverse:

$$f[n] = \frac{1}{N} \sum_{u=0}^{N-1} F[u] \exp\left(2\pi j \frac{un}{N}\right)$$





Fourier Transform



- Fourier transform stores the **magnitude** and **phase** at each frequency
 - Magnitude encodes how much signal there is at a particular frequency
 - Phase encodes spatial information (indirectly)
 - For mathematical convenience, this is often notated in terms of real and complex numbers

Amplitude: $A = \pm \sqrt{R(\omega)^2 + I(\omega)^2}$ Phase: $\phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$

Euler's formula: $e^{inx} = \cos(nx) + i \sin(nx)$



Name	Signal	Transform
impulse		$\delta(x) \Leftrightarrow 1$
shifted impulse		$\delta(x - u) \Leftrightarrow e^{-j\omega u}$
box filter		$\text{box}(x/a) \Leftrightarrow a\text{sinc}(a\omega)$
tent		$\text{tent}(x/a) \Leftrightarrow a\text{sinc}^2(a\omega)$
Gaussian		$G(x; \sigma) \Leftrightarrow \frac{\sqrt{2\pi}}{\sigma} G(\omega; \sigma^{-1})$
Laplacian of Gaussian		$(\frac{x^2}{\sigma^4} - \frac{1}{\sigma^2})G(x; \sigma) \Leftrightarrow -\frac{\sqrt{2\pi}}{\sigma} \omega^2 G(\omega; \sigma^{-1})$
Gabor		$\cos(\omega_0 x)G(x; \sigma) \Leftrightarrow \frac{\sqrt{2\pi}}{\sigma} G(\omega \pm \omega_0; \sigma^{-1})$
unsharp mask		$(1 + \gamma)\delta(x) - \gamma G(x; \sigma) \Leftrightarrow (1 + \gamma) - \frac{\sqrt{2\pi}\gamma}{\sigma} G(\omega; \sigma^{-1})$
windowed sinc		$\text{rcos}(x/(aW)) \text{sinc}(x/a) \Leftrightarrow (\text{see Figure 3.29})$



Today's Class



- Frequency domain and Fourier transform
- 1D Discrete Fourier Transform
- **2D Image Fourier Transform**
- Sampling
- Hybrid Image

For images, the 2D DFT

1D Discrete Fourier Transform (DFT) transforms a signal $f[n]$ into $F[u]$ as:

$$F[u] = \sum_{n=0}^{N-1} f[n] \exp \left(-2\pi j \frac{un}{N} \right)$$

2D Discrete Fourier Transform (DFT) transforms an image $f[n,m]$ into $F[u,v]$ as:

$$F[u, v] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f[n, m] \exp \left(-2\pi j \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$

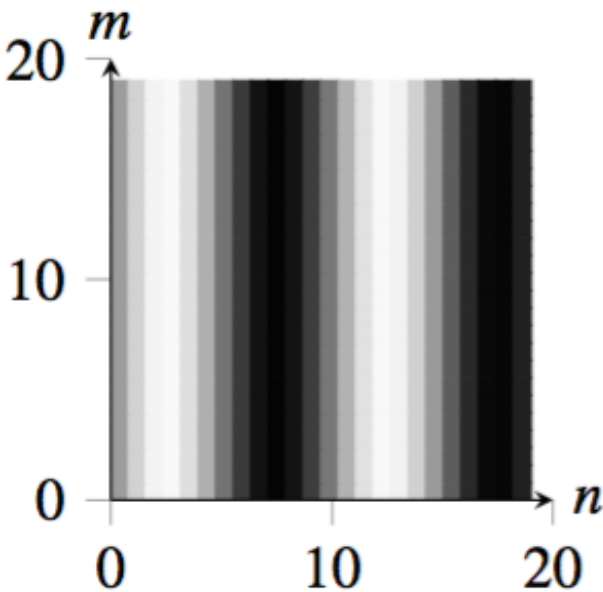


Waves in 2D

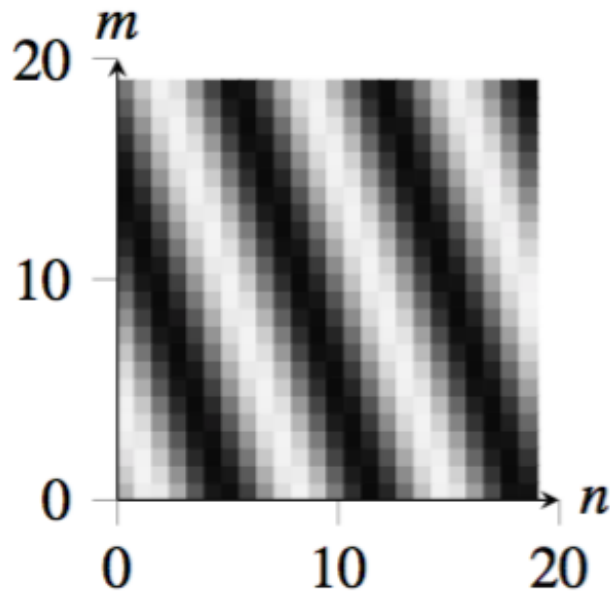


$$s_{u,v}[n,m] = A \sin \left(2\pi \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$

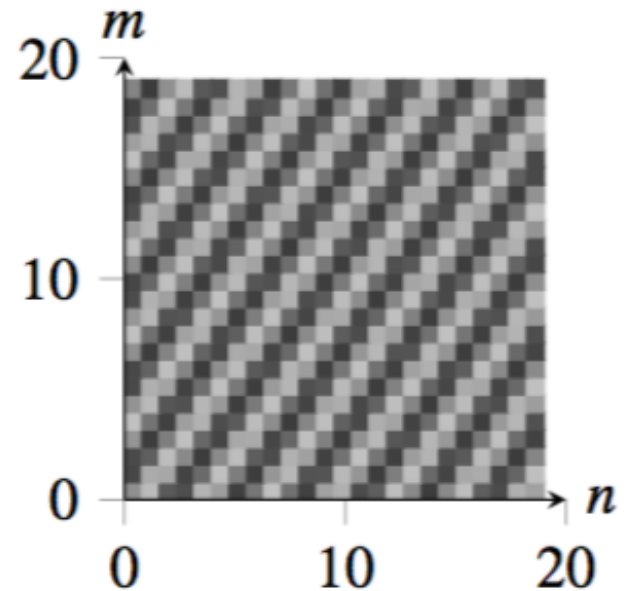
$$c_{u,v}[n,m] = A \cos \left(2\pi \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$



$$u = 2, v = 0$$



$$u = 3, v = 1$$

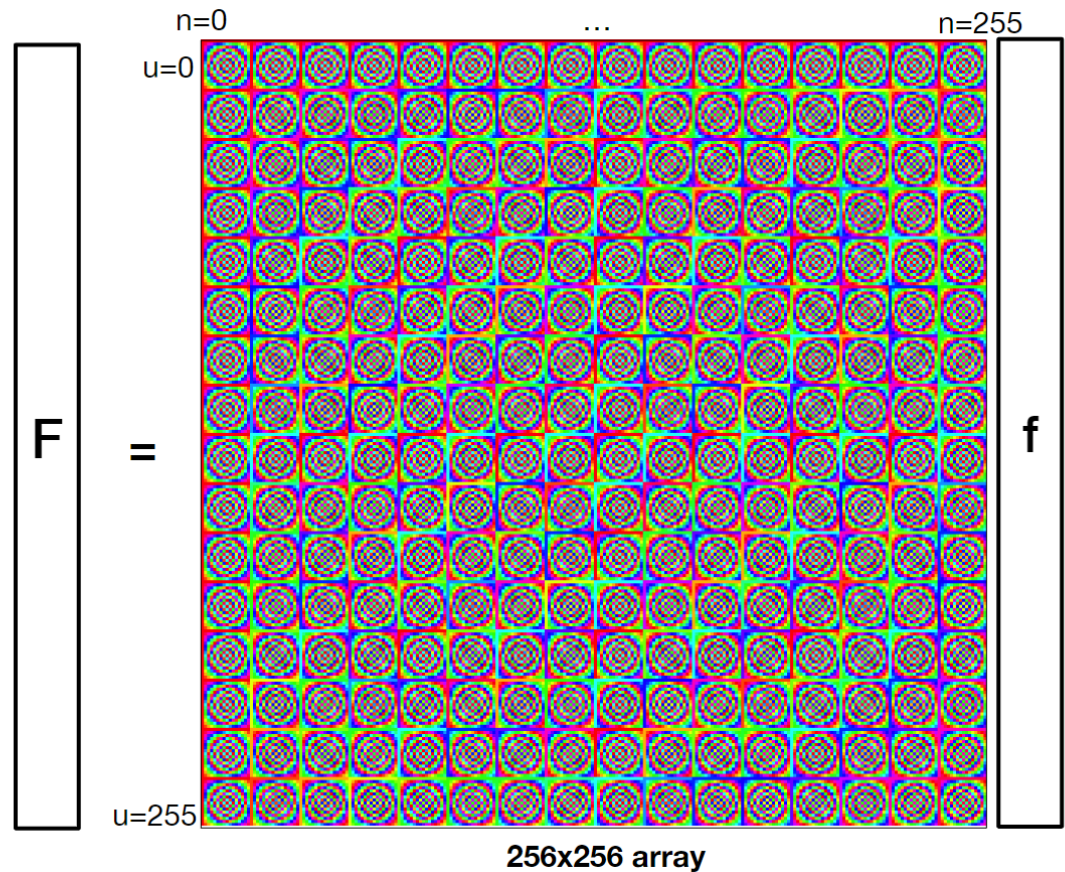


$$u = 7, v = -5$$

Visualizing the 2D DFT coefficients

$$F[u, v] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f[n, m] \exp \left(-2\pi j \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$

For N=M=16



Visualizing the image Fourier transform

$$F[u, v] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f[n, m] \exp \left(-2\pi j \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$

The values of $F[u, v]$ are complex.

Using the real and imaginary components:

$$F[u, v] = \text{Re}\{F[u, v]\} + j \text{Imag}\{F[u, v]\}$$

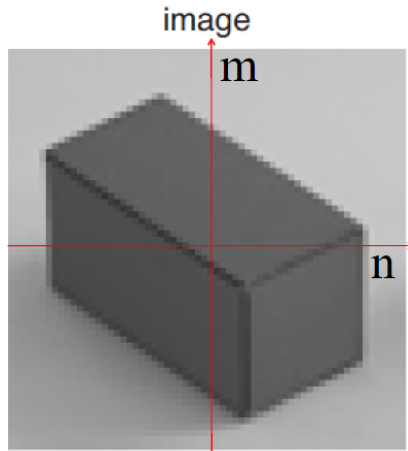
Or using a polar decomposition:

$$F[u, v] = A[u, v] \exp(j\theta[u, v])$$

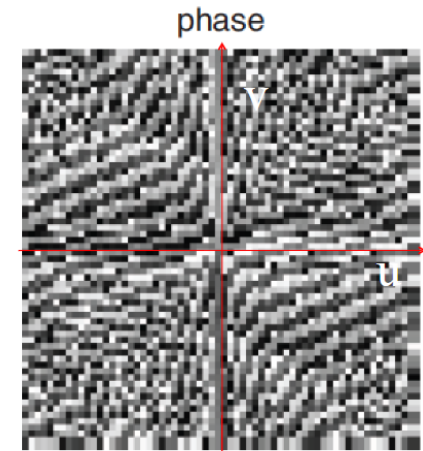
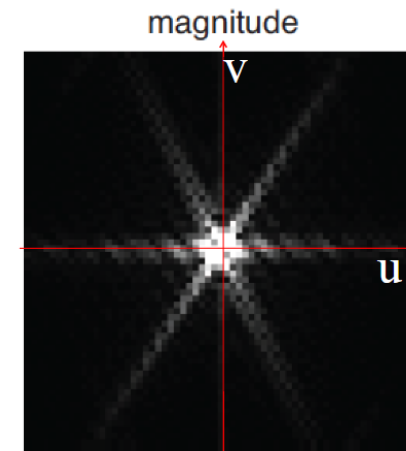
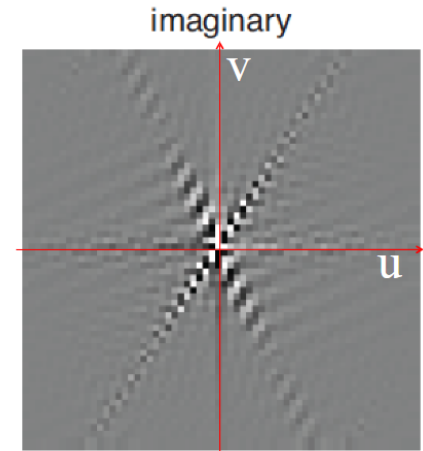
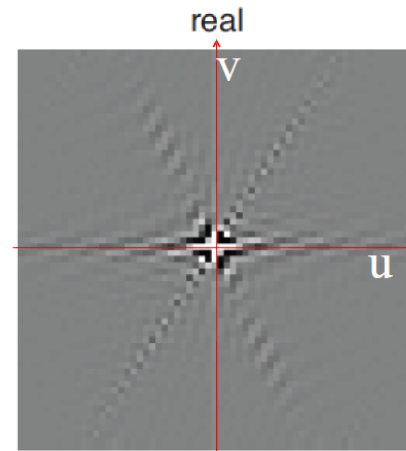
Amplitude Phase

Visualizing the image Fourier transform

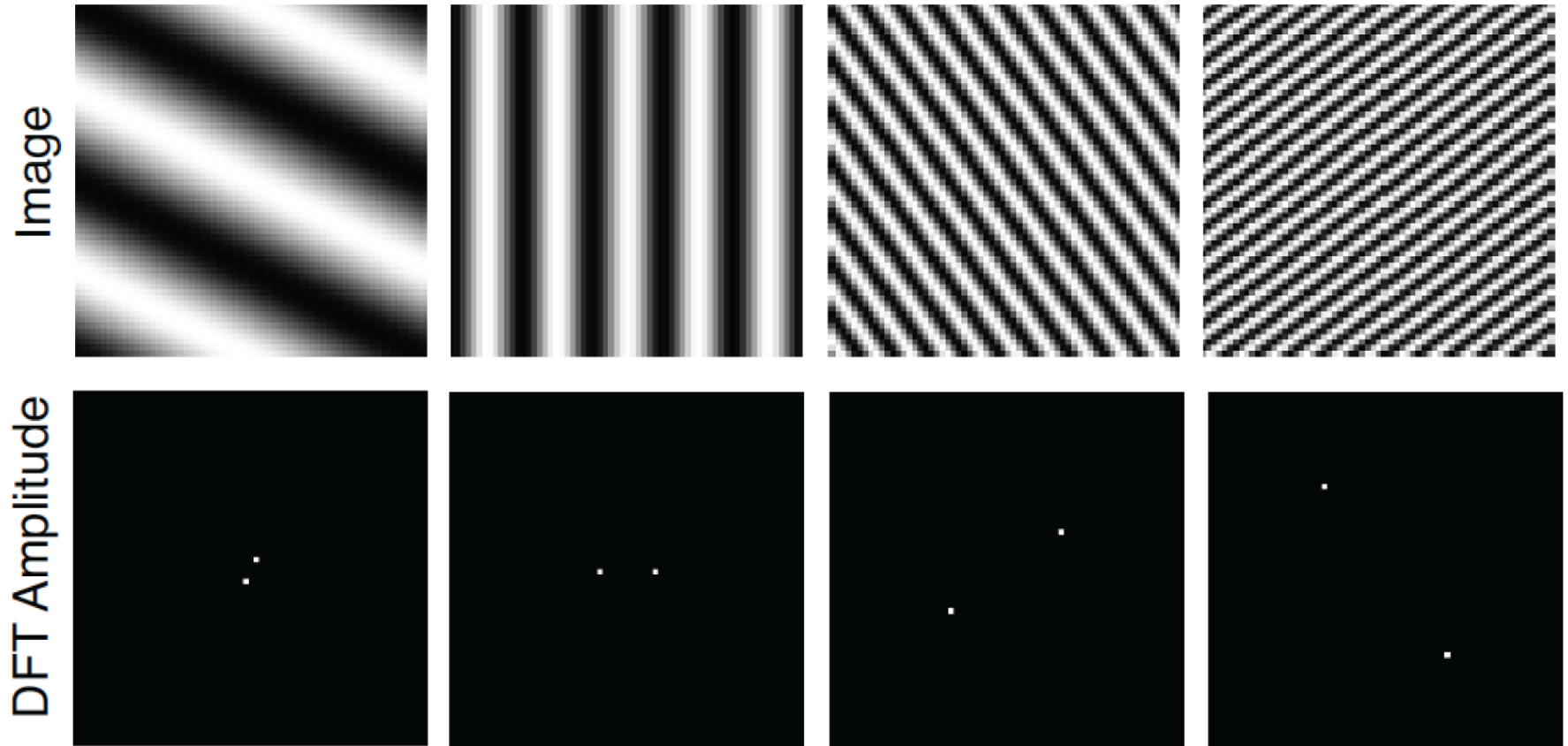
$f[n, m]$



$F[u, v]$

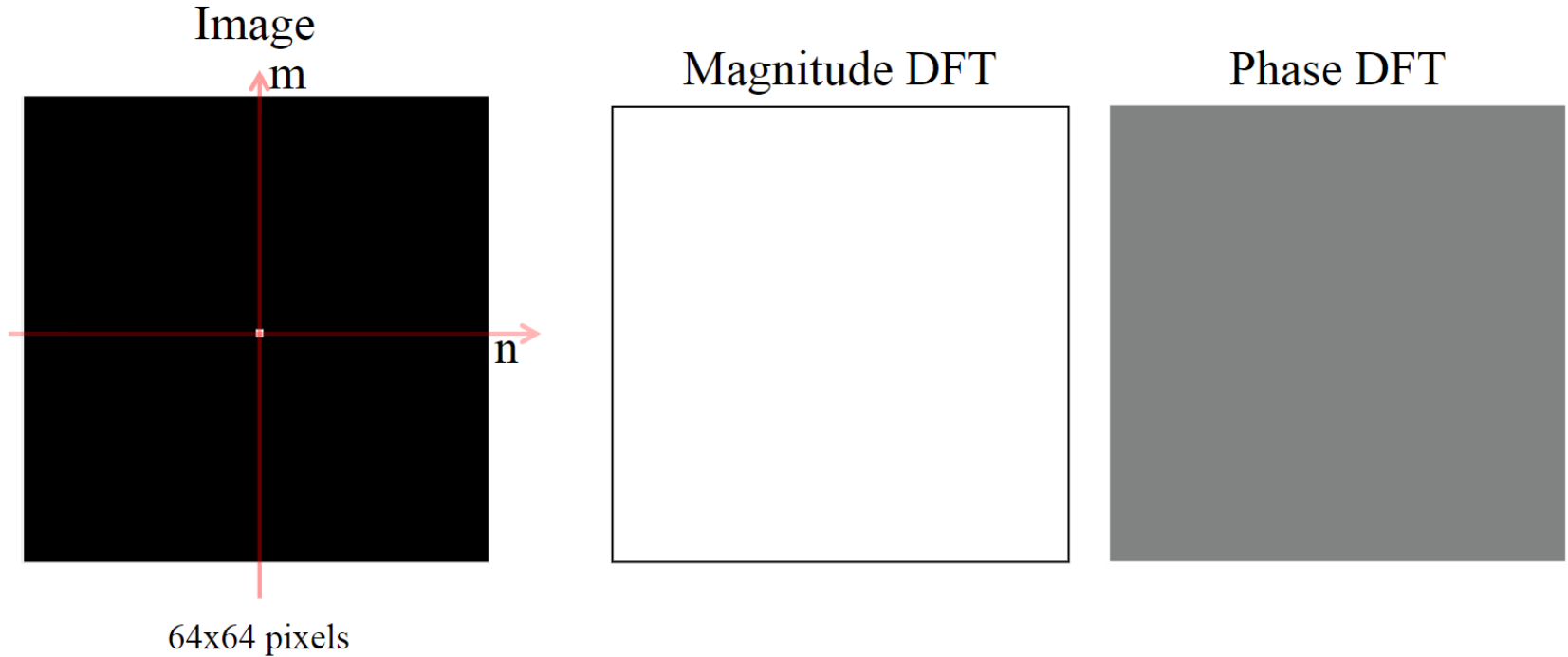


Simple Fourier transforms

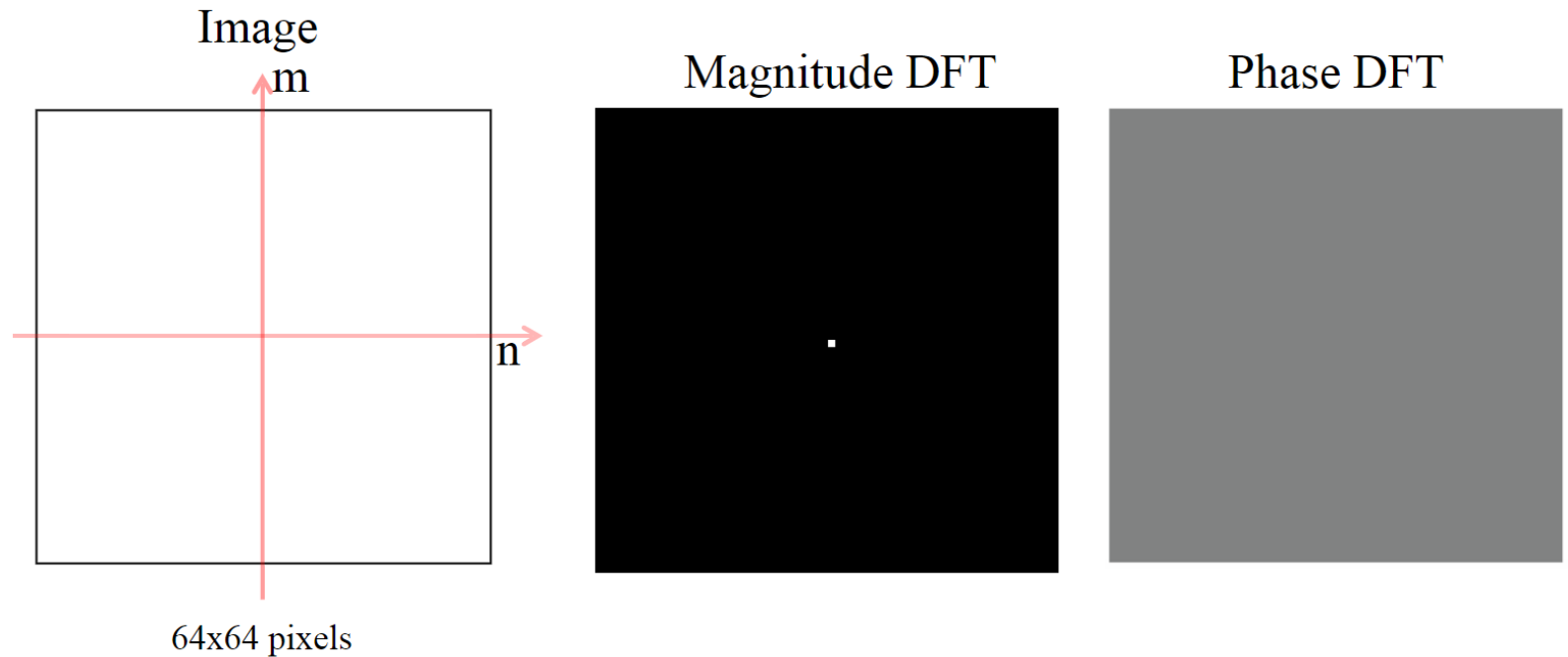


Images are 64x64 pixels. The wave is a cosine, therefore DFT phase is zero.

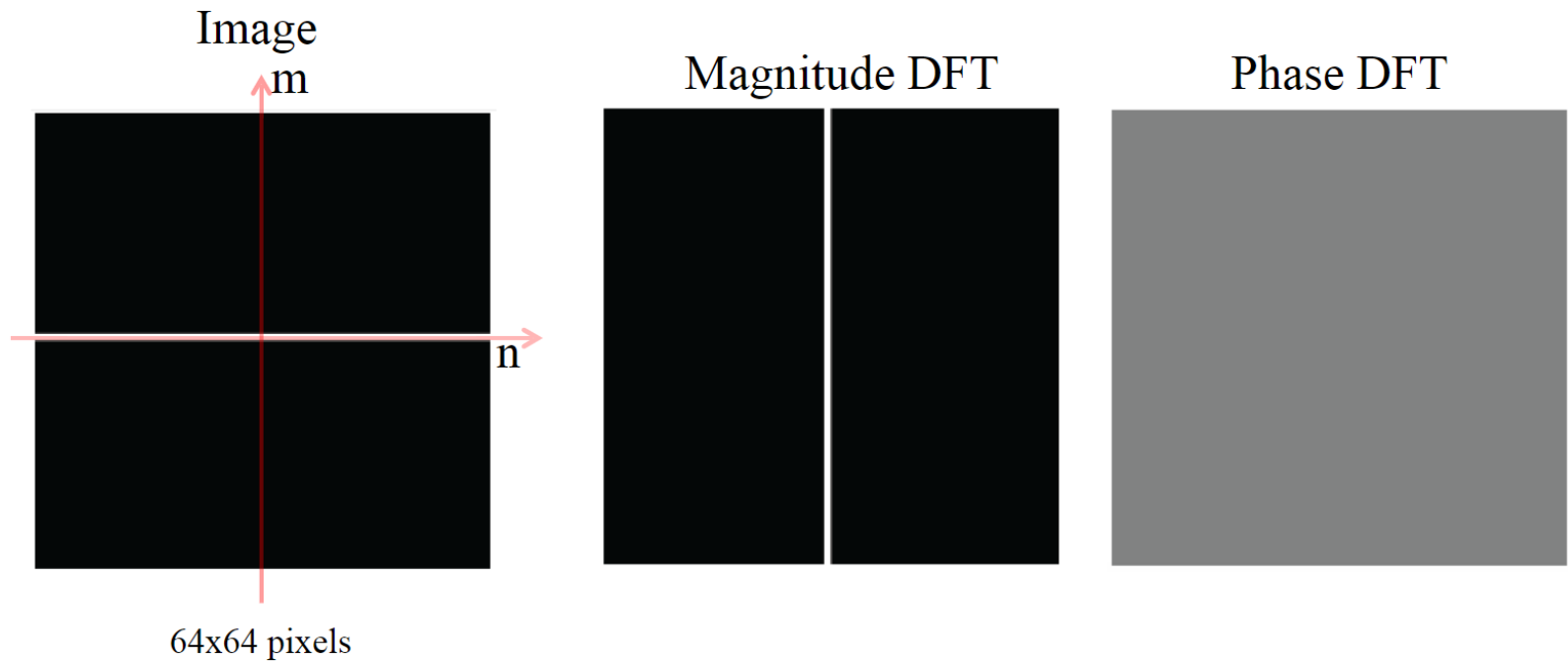
Some important Fourier transforms



Some important Fourier transforms



Some important Fourier transforms

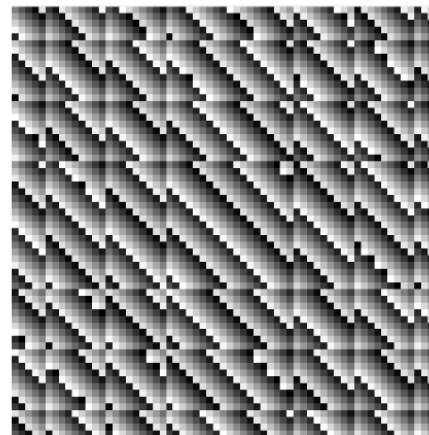
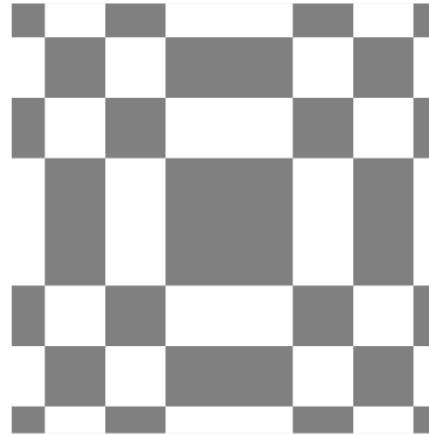


Some important Fourier transforms

Image

Magnitude DFT

Phase DFT



Translation

Shifts of an image only produce changes on the phase of the DFT.

Some important Fourier transforms

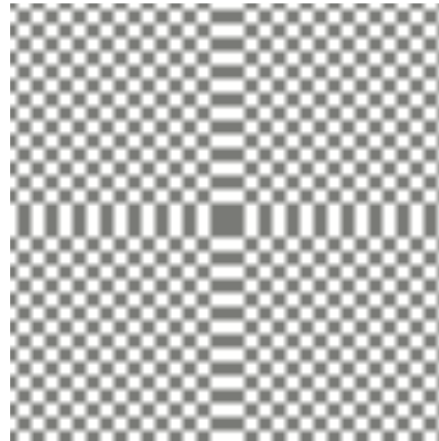
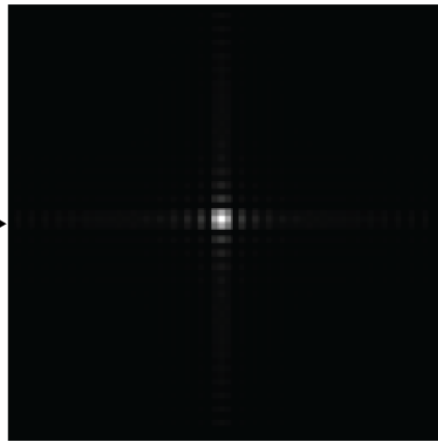
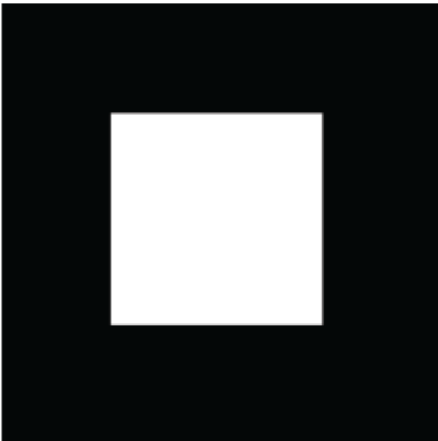
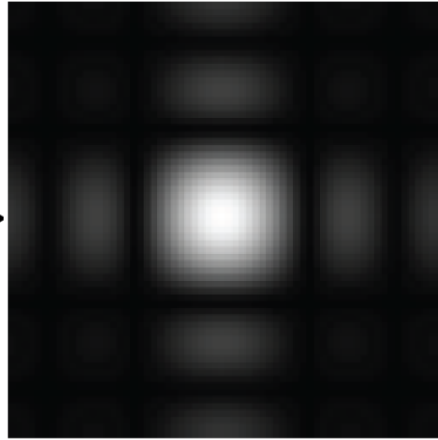
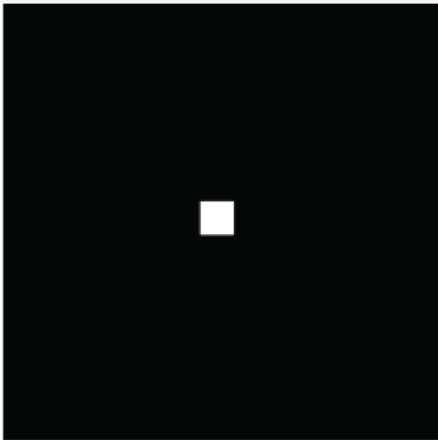
Image

Magnitude DFT

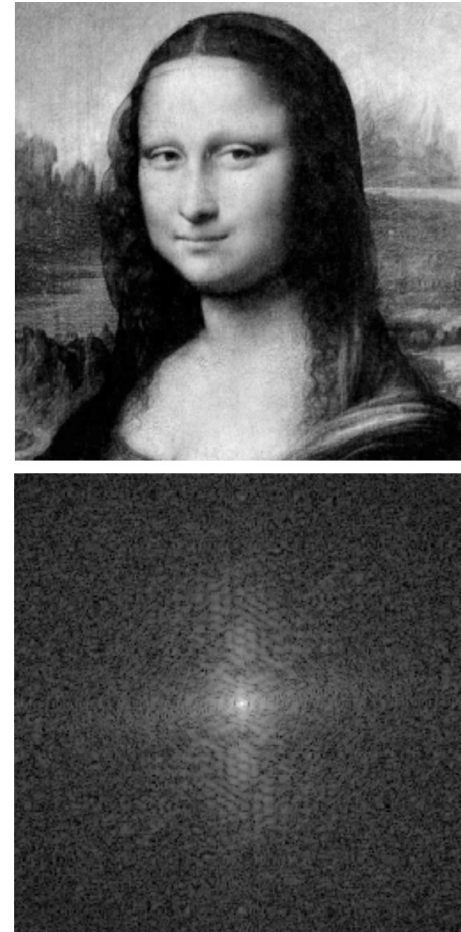
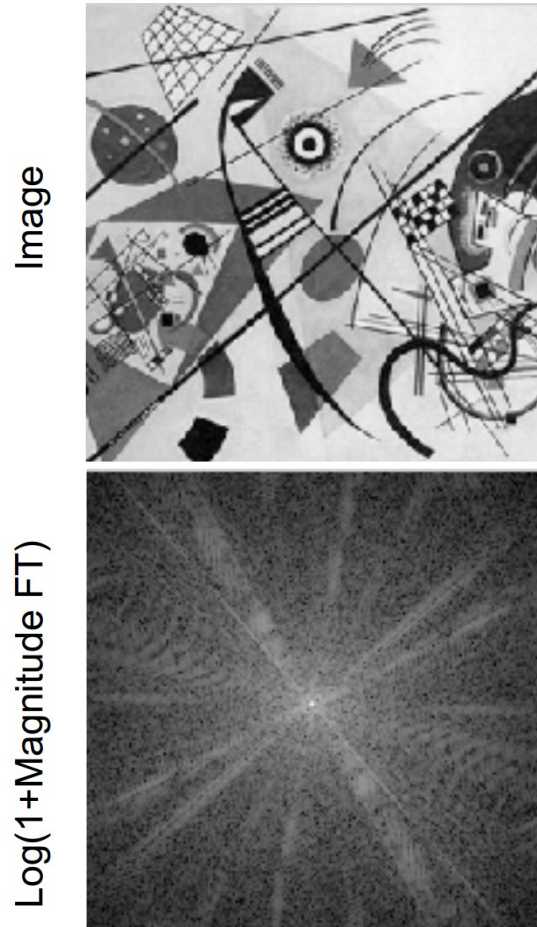
Phase DFT

Scale

Small image details
produce content in high
spatial frequencies



The Fourier Transform of some important images



73



Question



Match the spatial domain image to the Fourier magnitude image

1 2 3 4 5

B

A

C

D

E

The image contains five Fourier magnitude images (1-5) and five spatial domain images (A-E).
1: A single bright spot in the center of a dark blue square.
2: A bright, circular, slightly blurred spot in the center of a dark blue square.
3: Two bright, circular, slightly blurred spots side-by-side in the center of a dark blue square.
4: A bright, vertical, slightly blurred line in the center of a dark blue square.
5: A bright, cross-shaped, slightly blurred pattern in the center of a dark blue square.
A: A small 3x3 grayscale pattern with a white center and dark corners.
B: A grayscale image of a field of white tulips.
C: A grayscale image of a traditional Chinese building with a tiled roof.
D: A grayscale image of a bright, circular, slightly blurred spot in the center of a dark square.
E: A grayscale image of a group of people standing in shallow water near a rocky shore.

The inverse Discrete Fourier transform

2D Discrete Fourier Transform (DFT) transforms an image $f[n, m]$ into $F[u, v]$ as:

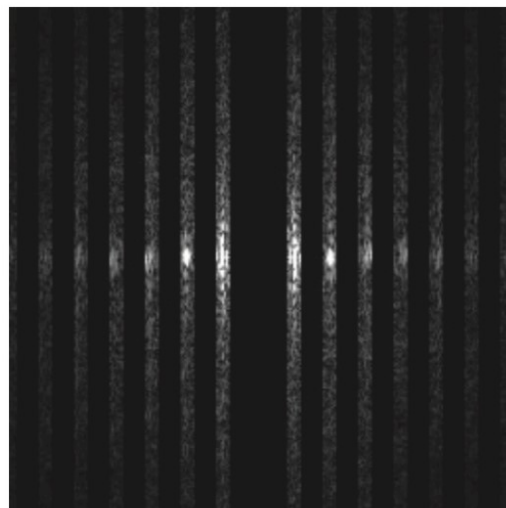
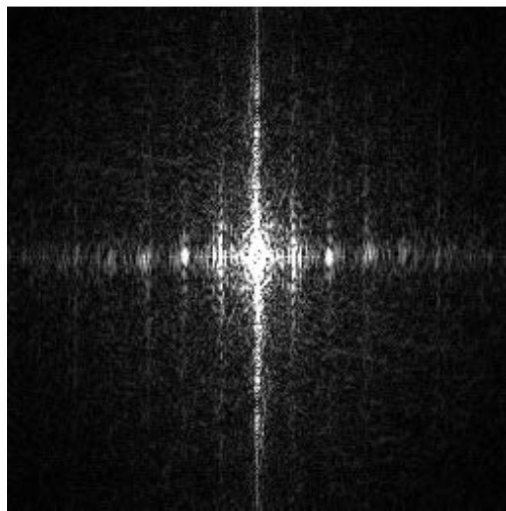
$$F[u, v] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f[n, m] \exp \left(-2\pi j \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$

The inverse of the 2D DFT is:

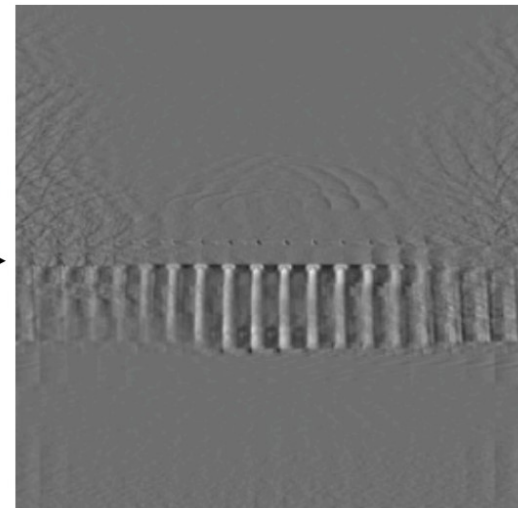
$$f[n, m] = \frac{1}{NM} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} F[u, v] \exp \left(+2\pi j \left(\frac{un}{N} + \frac{vm}{M} \right) \right)$$



DFT
→

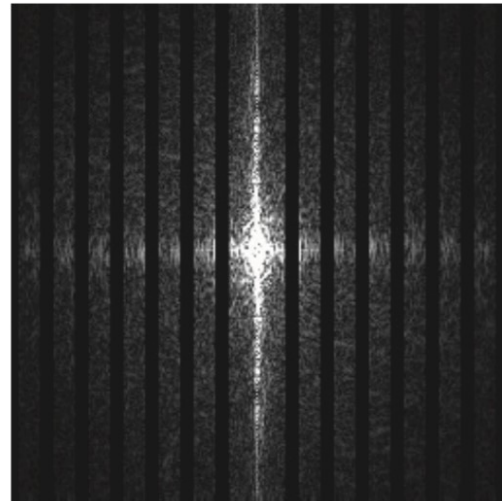
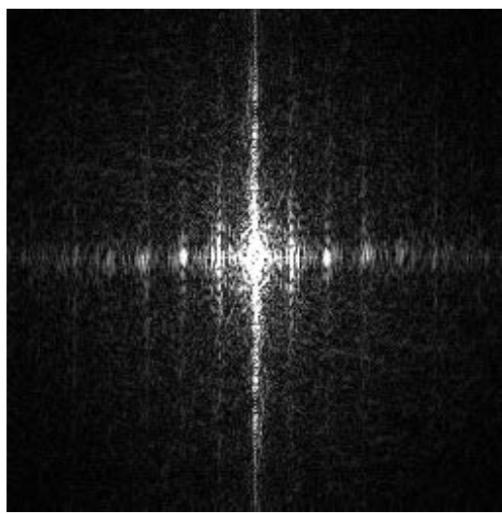


DFT⁻¹
→





DFT
→



DFT^{-1}
→





Questions

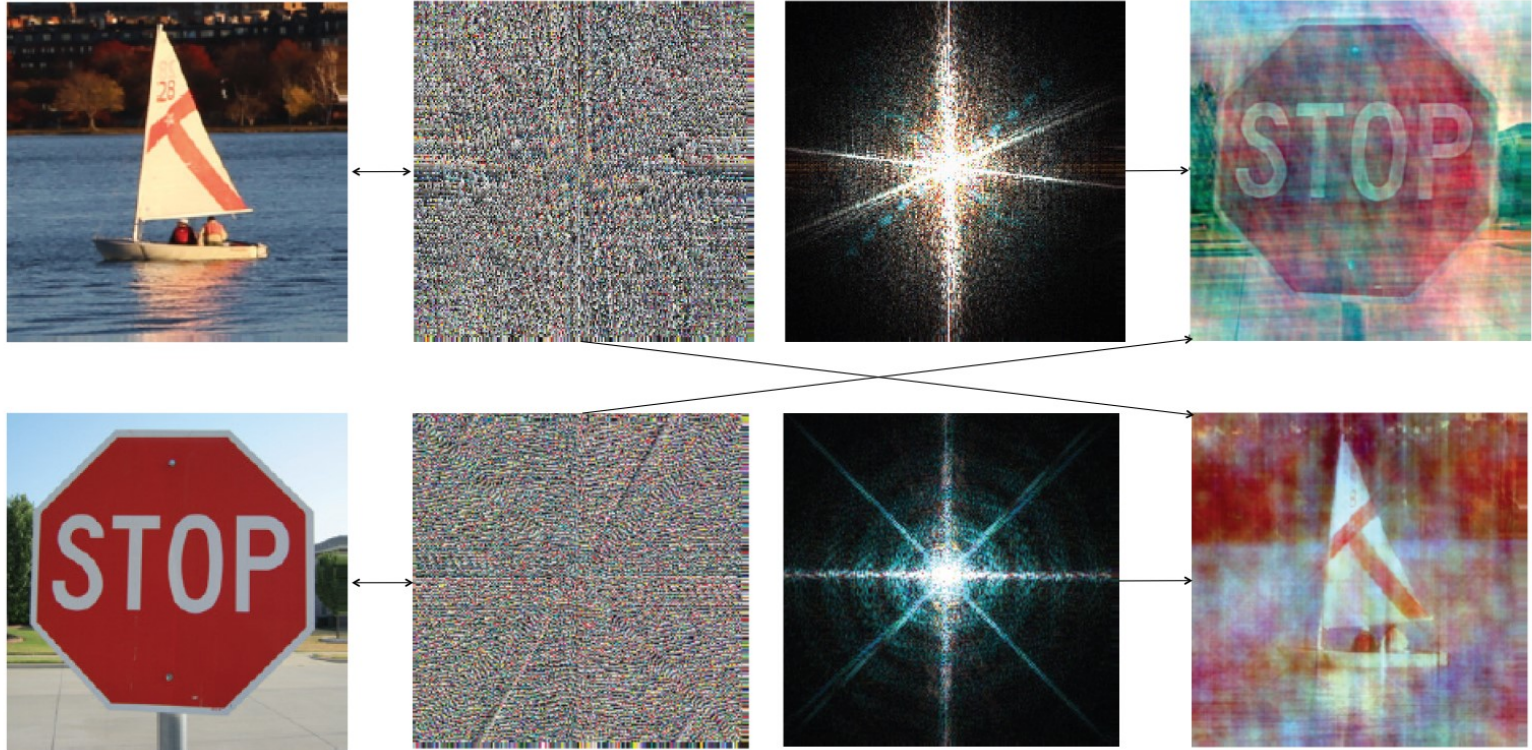


Which has more information, the phase or the magnitude?

What happens if you take the phase from one image and combine it with the magnitude from another image?

Phase and Magnitude

$$F[u, v] = A[u, v] \exp(j\theta[u, v])$$

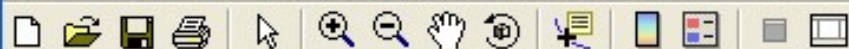


Each color channel is processed in the same way.

Figure 1



File Edit View Insert Tools Desktop Window Help



x 原图



y 原图



x 幅度谱 与 y 相位谱



y 幅度谱 与 x 相位谱





Properties of Fourier Transforms



- Linearity $\mathcal{F}[ax(t) + by(t)] = a\mathcal{F}[x(t)] + b\mathcal{F}[y(t)]$
- Fourier transform of a real signal is symmetric about the origin
- The energy of the signal is the same as the energy of its Fourier transform



The Convolution Theorem



- The Fourier transform of the convolution of two functions is the product of their Fourier transforms

$$F[g * h] = F[g]F[h]$$

- The inverse Fourier transform of the product of two Fourier transforms is the convolution of the two inverse Fourier transforms

$$F^{-1}[gh] = F^{-1}[g] * F^{-1}[h]$$

- **Convolution** in spatial domain is equivalent to **multiplication** in frequency domain!

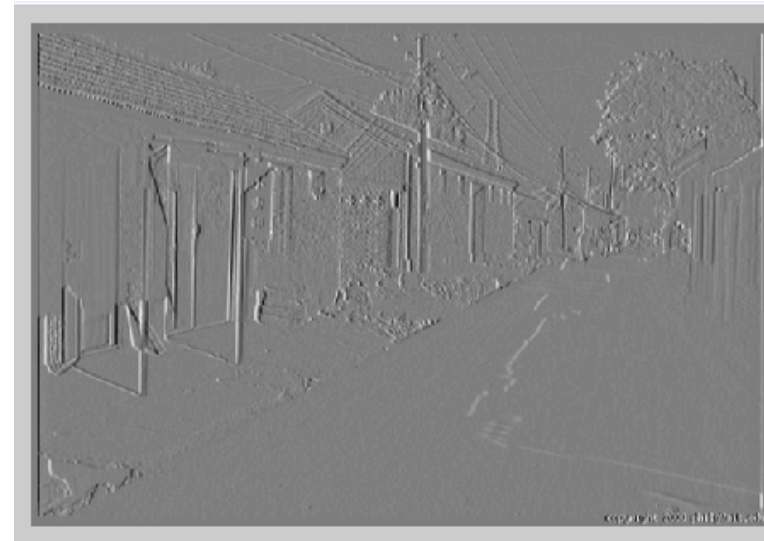


Filtering in spatial domain



1	0	-1
2	0	-2
1	0	-1

intensity image





Filtering in frequency domain



FFT



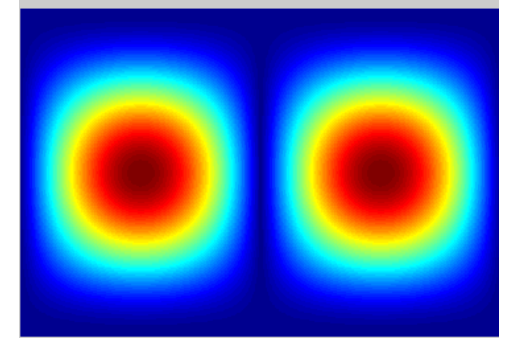
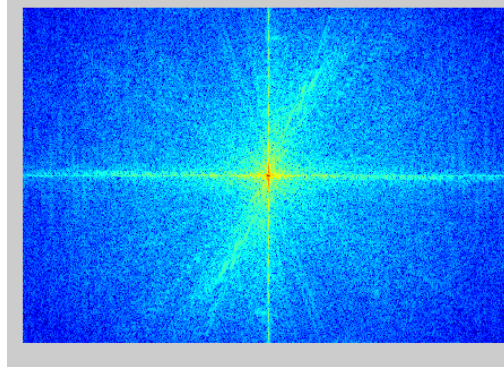
intensity image



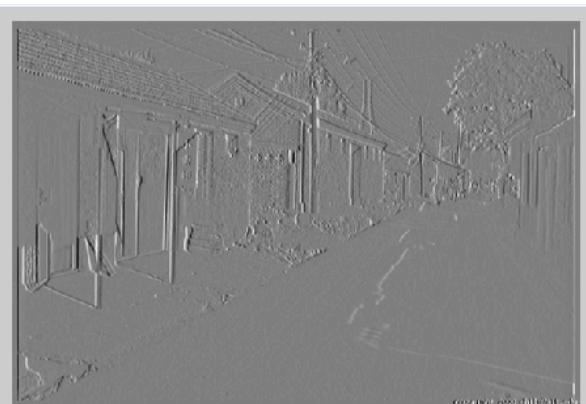
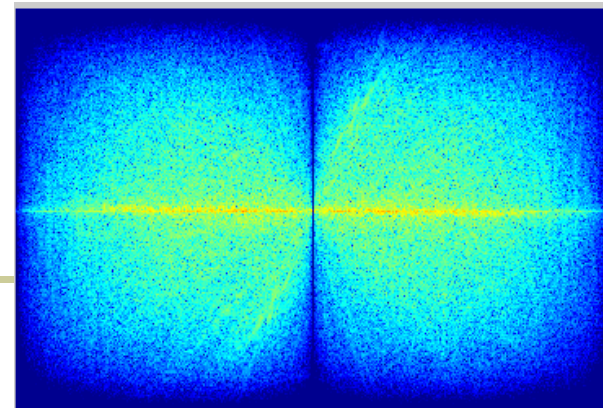
FFT



log fft magnitude



Inverse FFT



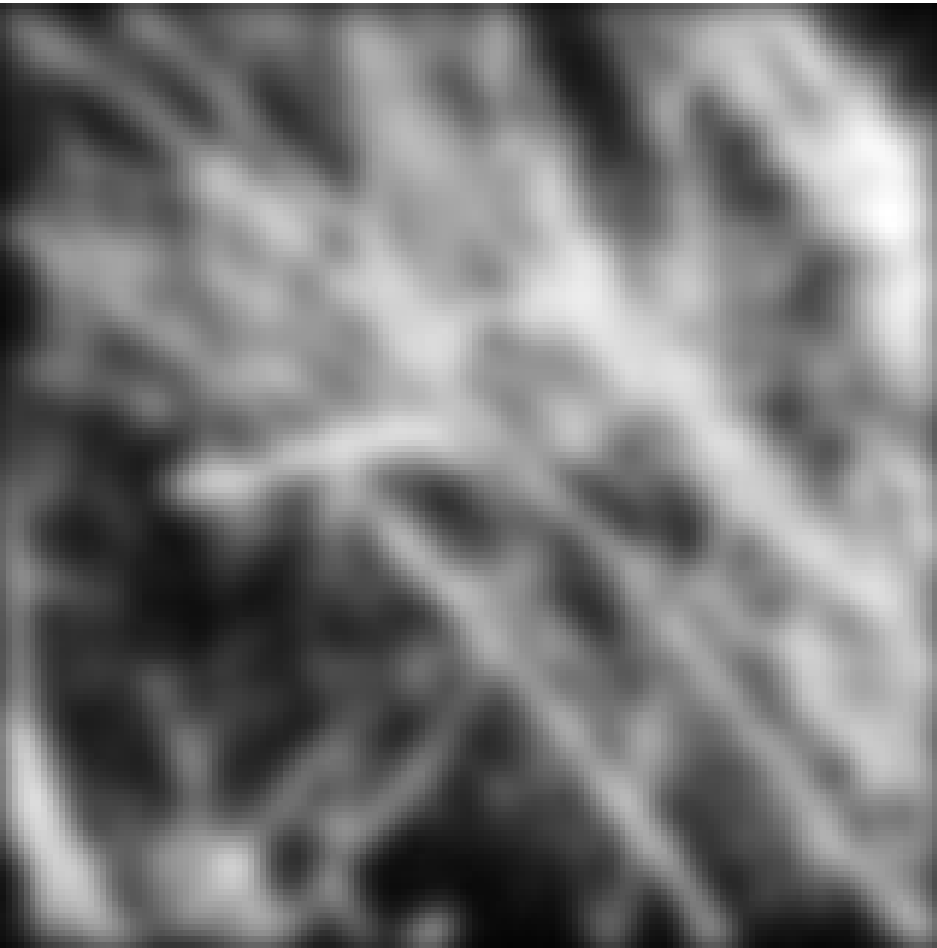


Filtering

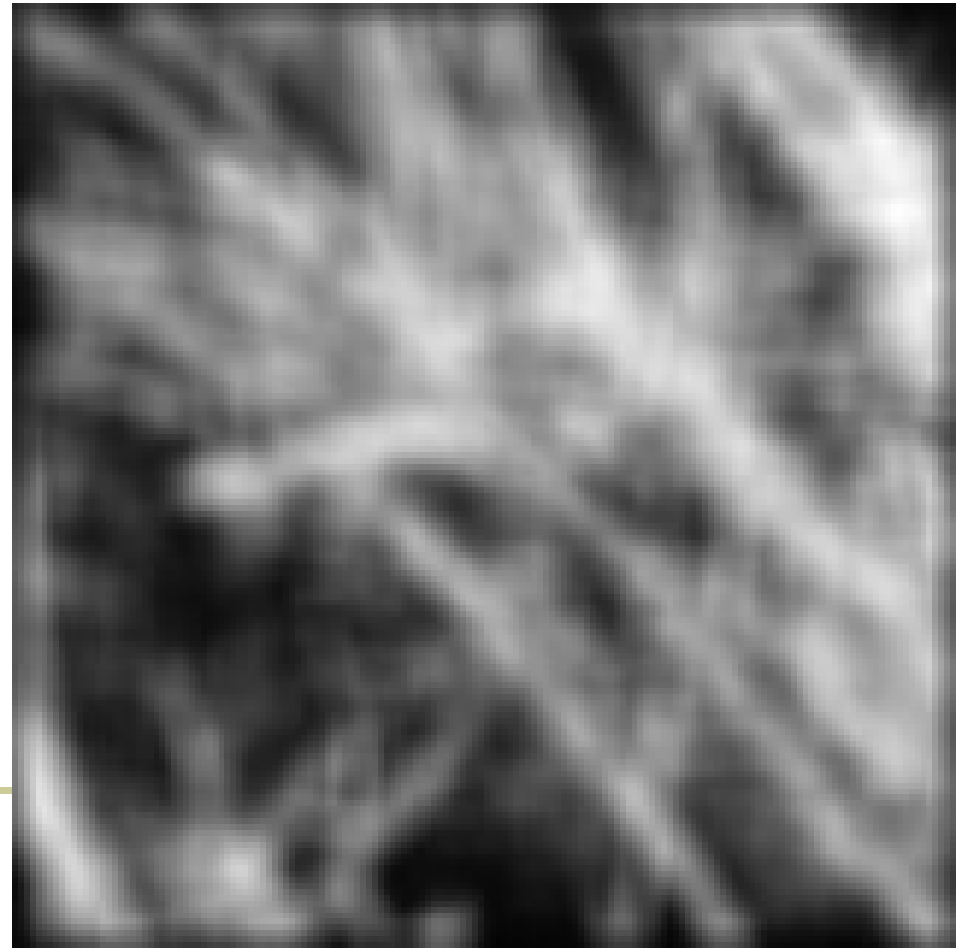


Why does the Gaussian give a nice smooth image, but the square filter give edgy artifacts?

Gaussian



Box filter





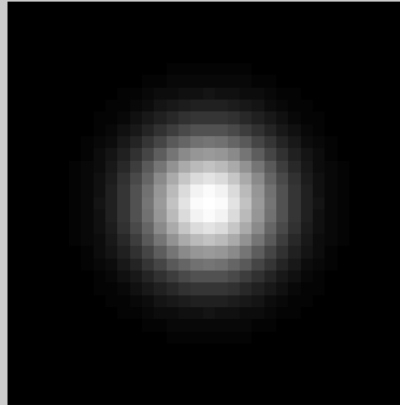
Gaussian Filter



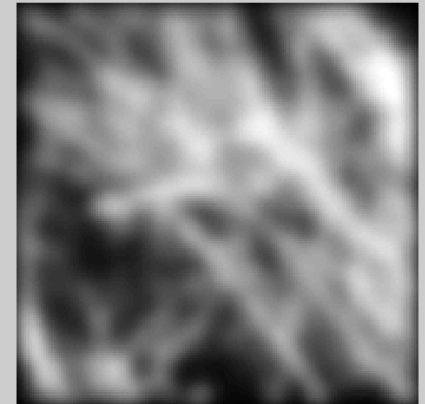
intensity image



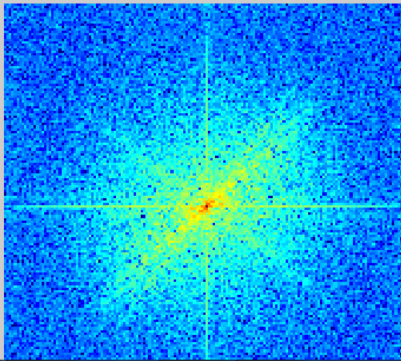
filter: gaussian



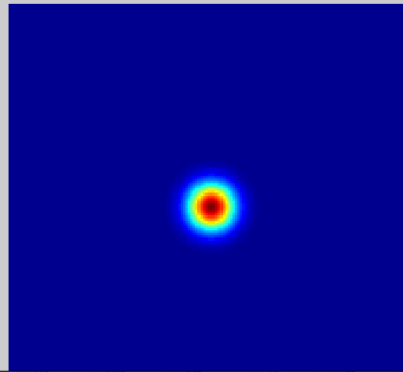
filtered image



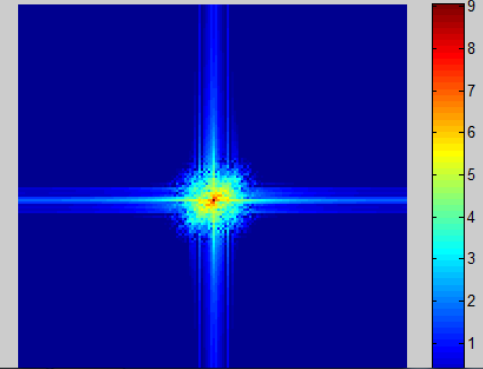
log ft magnitude of image



filter: gaussian

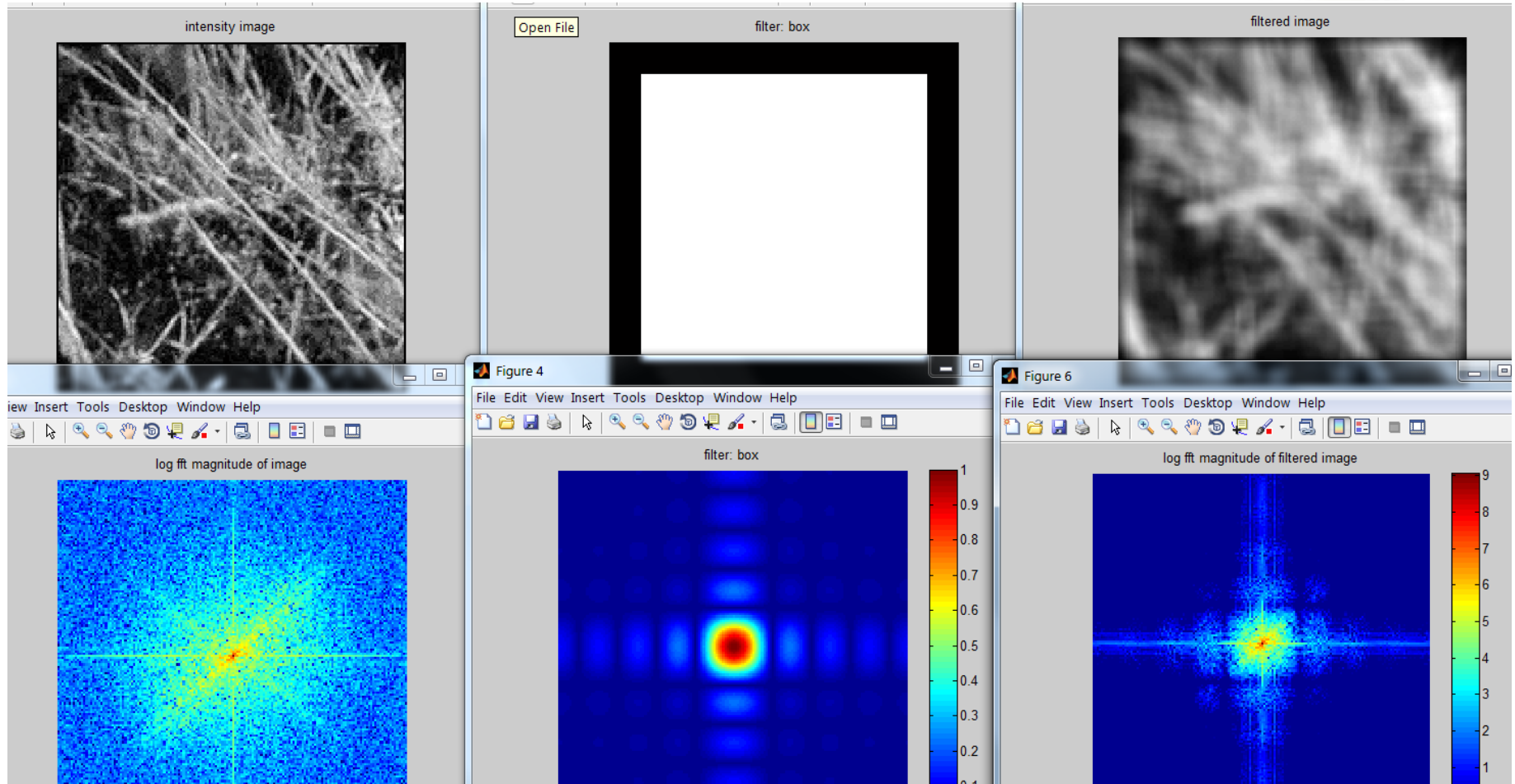


log ft magnitude of filtered image





Box Filter





Today's Class



- Frequency domain and Fourier transform
- 1D Discrete Fourier Transform
- 2D Image Fourier Transform
- **Sampling**
- Hybrid Image



Sampling

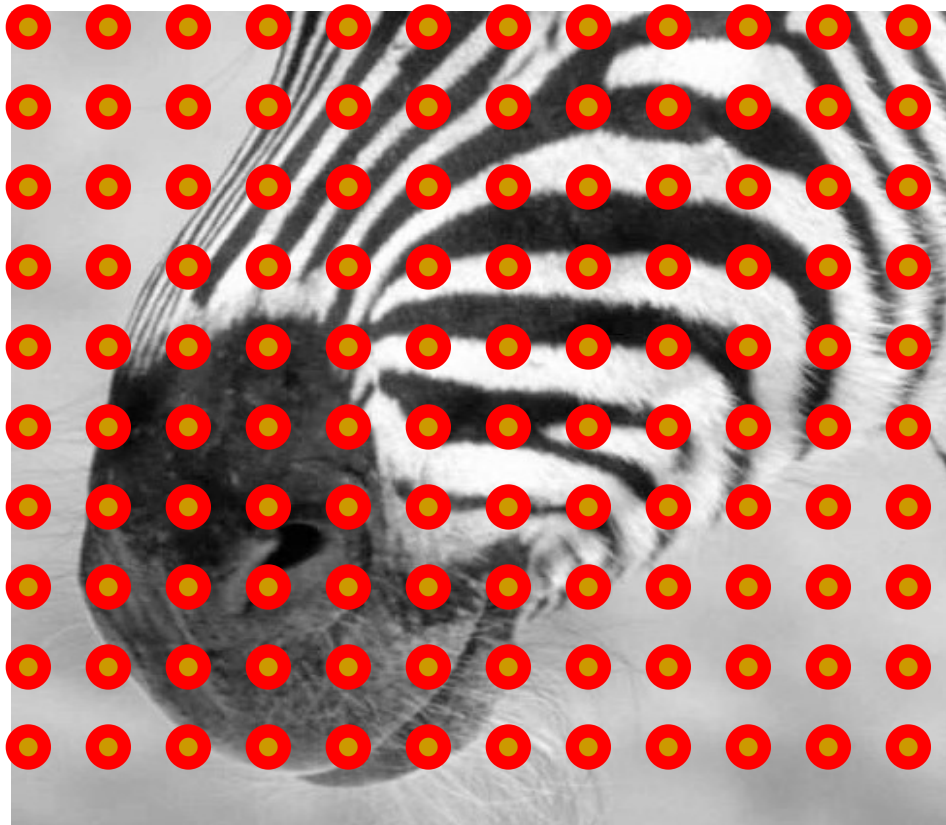


Why does a lower resolution image still make sense to us? What do we lose?





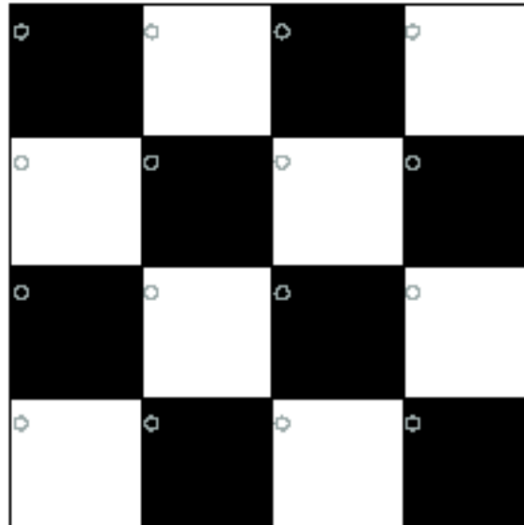
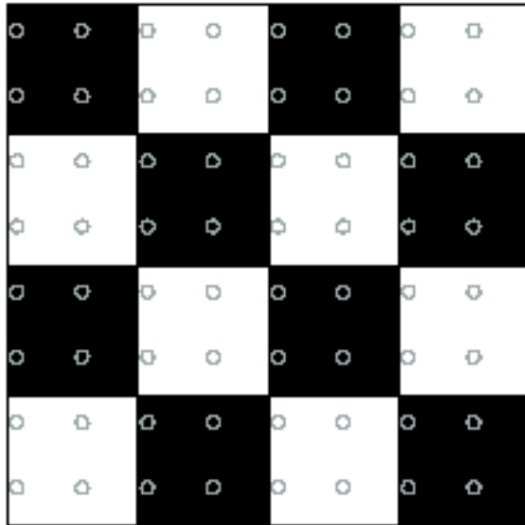
Subsampling by a factor of 2



Throw away every other row and column
to create a 1/2 size image

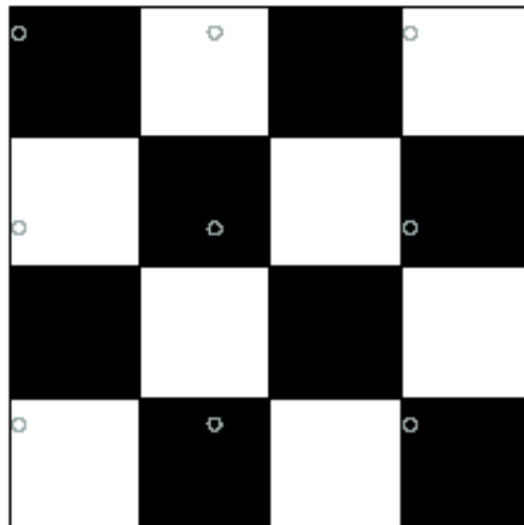
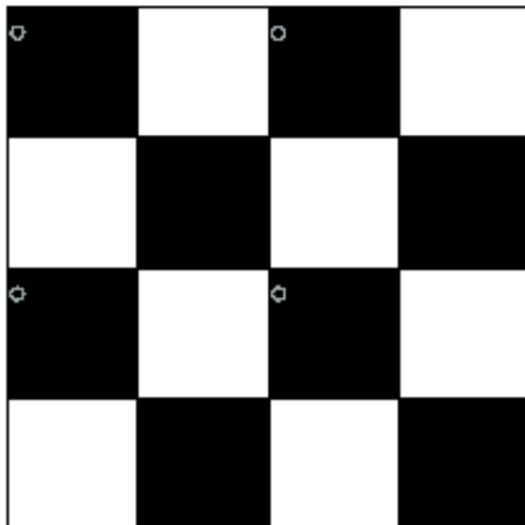


How should we go about sampling



Let's resample the checkerboard by taking one sample at each circle.

In the top left board, the new representation is reasonable. Top right also yields a reasonable representation.



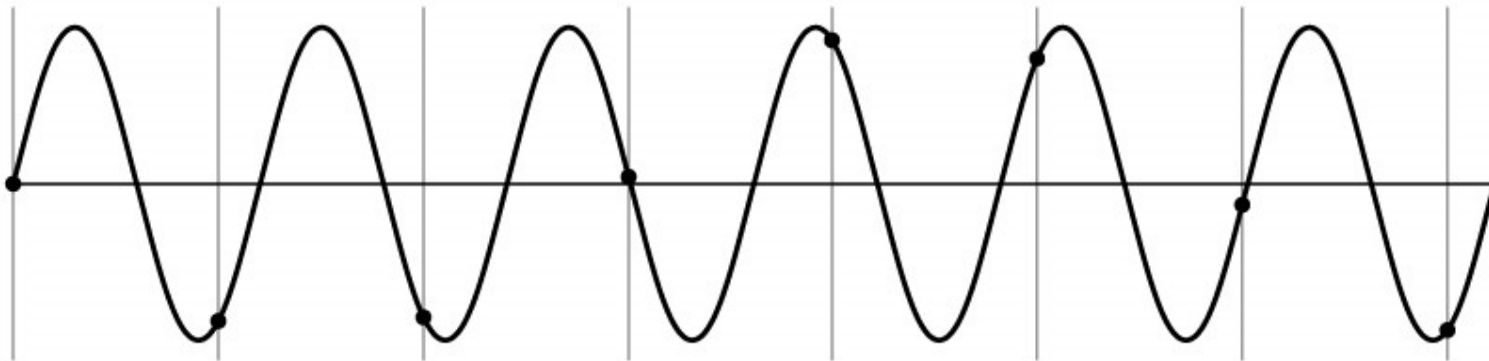
Bottom left is all black (dubious) and bottom right has checks that are too big.



How should we go about sampling



- 1D example (sinewave):

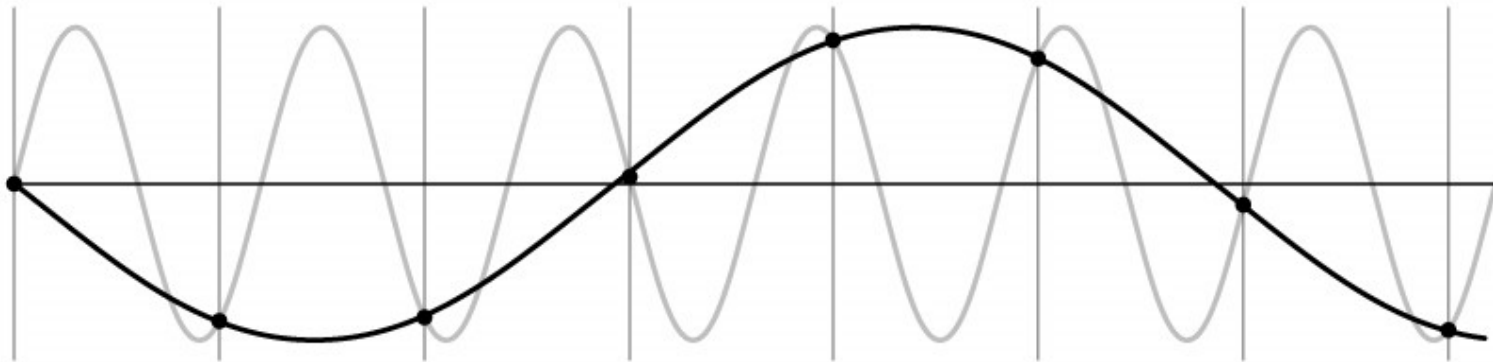




How should we go about sampling



- 1D example (sinewave):

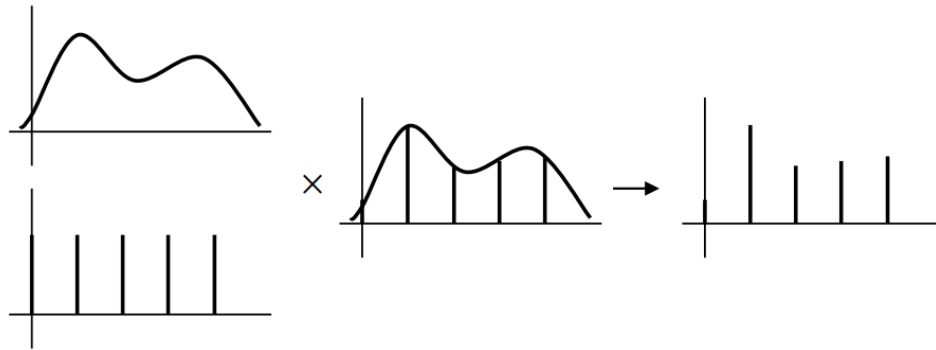




Fourier Interpretation: Sampling



- Sampling in the spatial domain is like multiplying with a spike function.



- Sampling in the frequency domain is like...

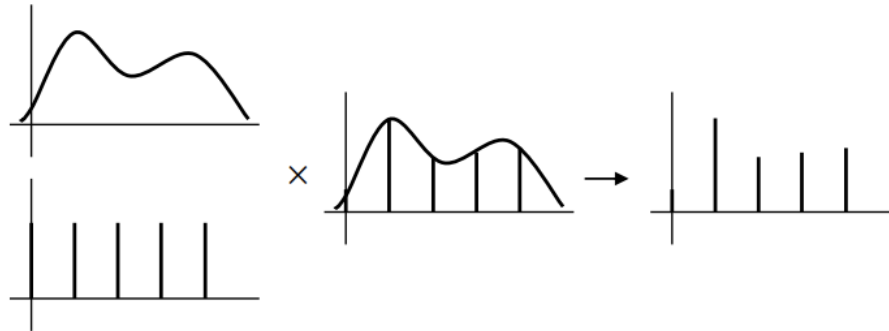
?



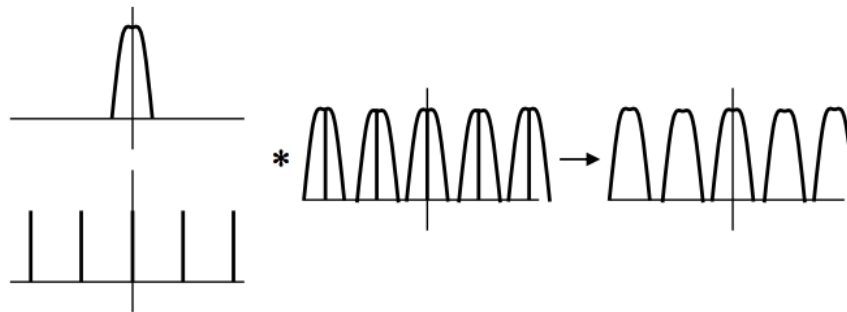
Fourier Interpretation: Sampling



- Sampling in the spatial domain is like multiplying with a spike function.

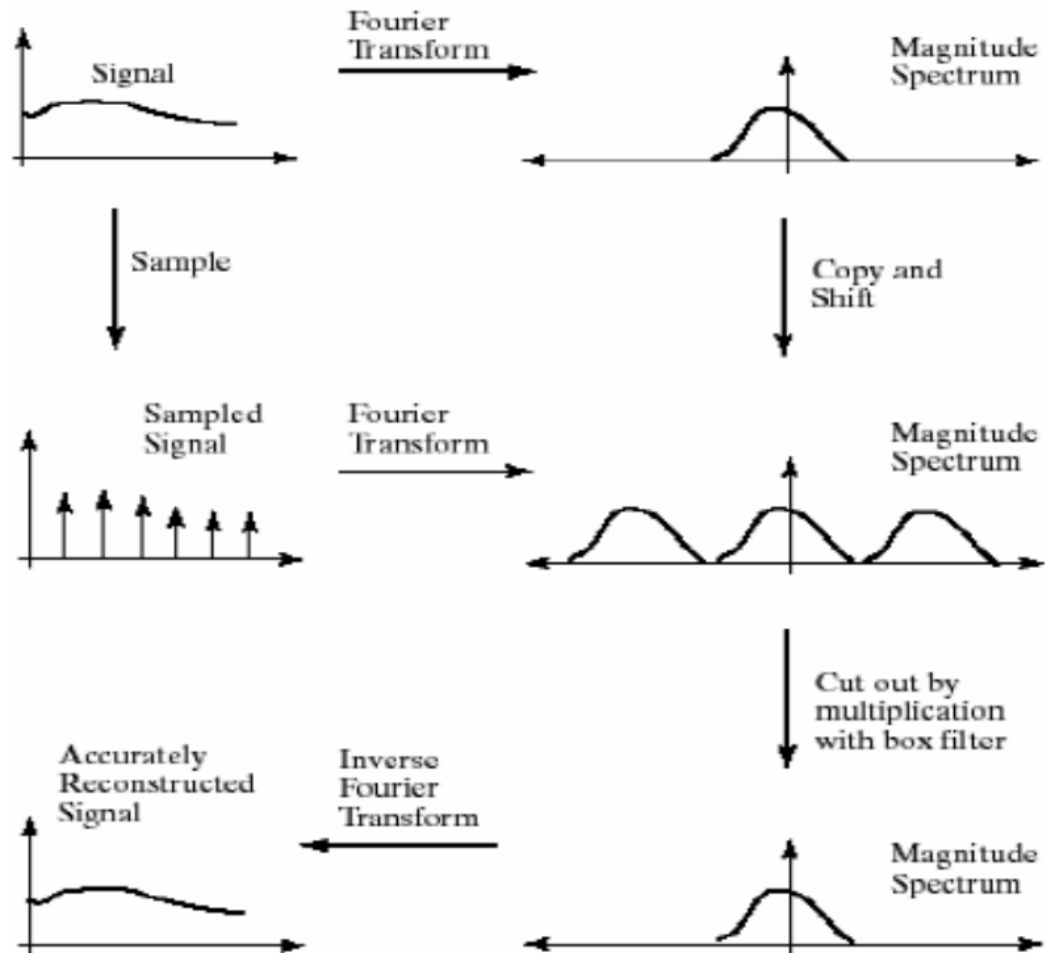


- Sampling in the frequency domain is like convolving with a spike function.



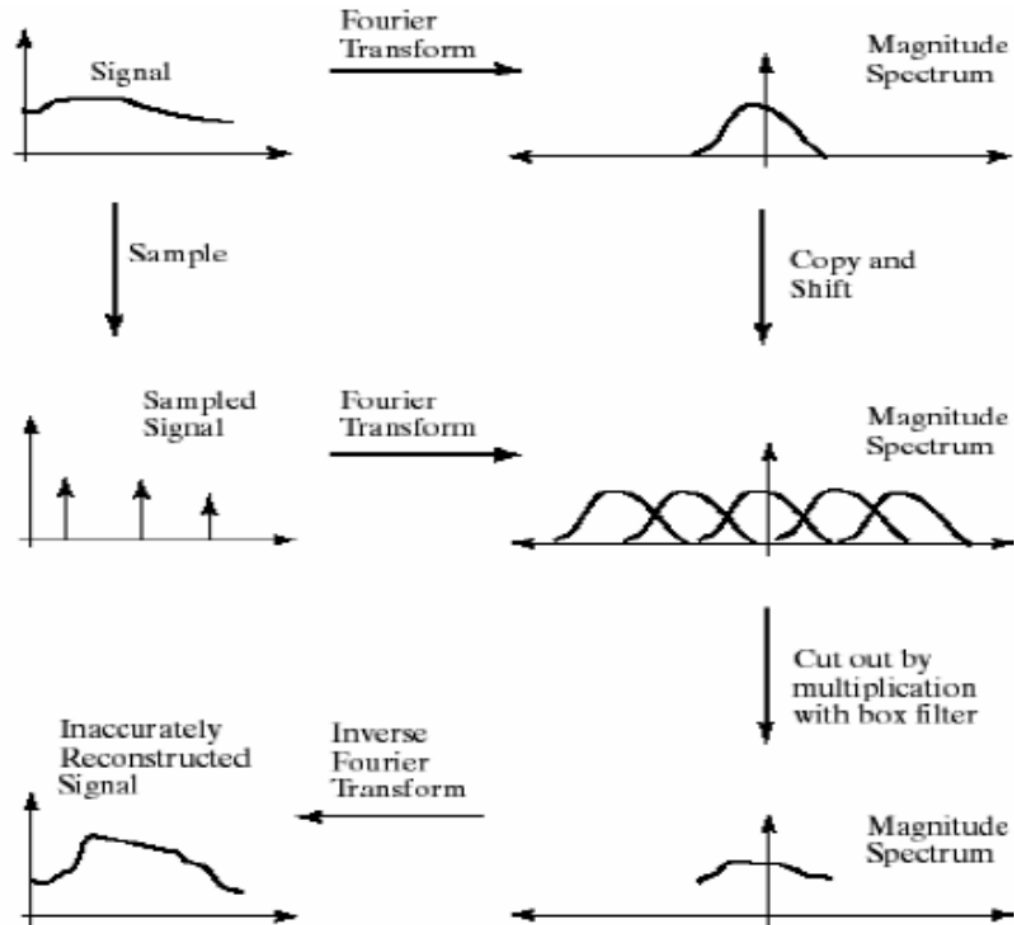


Fourier Interpretation: Sampling





Fourier Interpretation: Sampling





Aliasing problem (混叠)



- Sub-sampling may be dangerous....
- Characteristic errors may appear:
 - “Wagon wheels rolling the wrong way in movies”
 - “Checkerboards disintegrate in ray tracing”
 - “Striped shirts look funny on color television”

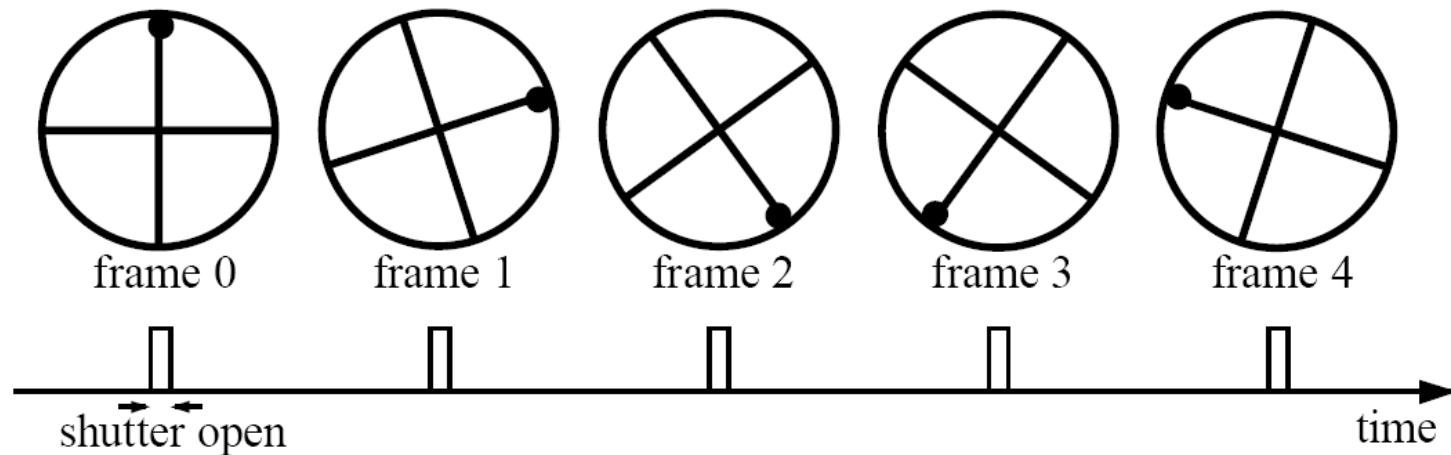


Aliasing in video



Imagine a spoked wheel moving to the right (rotating clockwise).
Mark wheel with dot so we can see what's happening.

If camera shutter is only open for a fraction of a frame time (frame time = $1/30$ sec. for video, $1/24$ sec. for film):



Without dot, wheel appears to be rotating slowly backwards!
(counterclockwise)



Sampling and aliasing



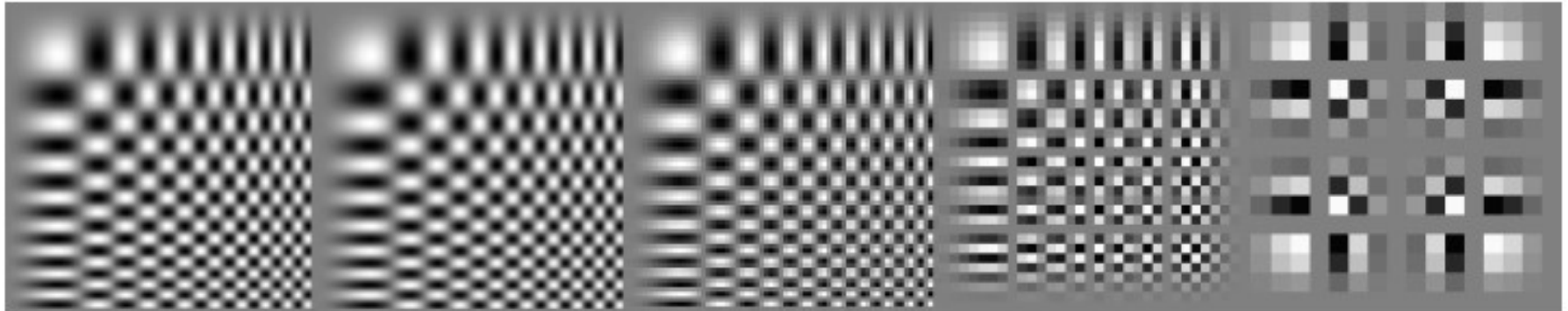
256x256

128x128

64x64

32x32

16x16

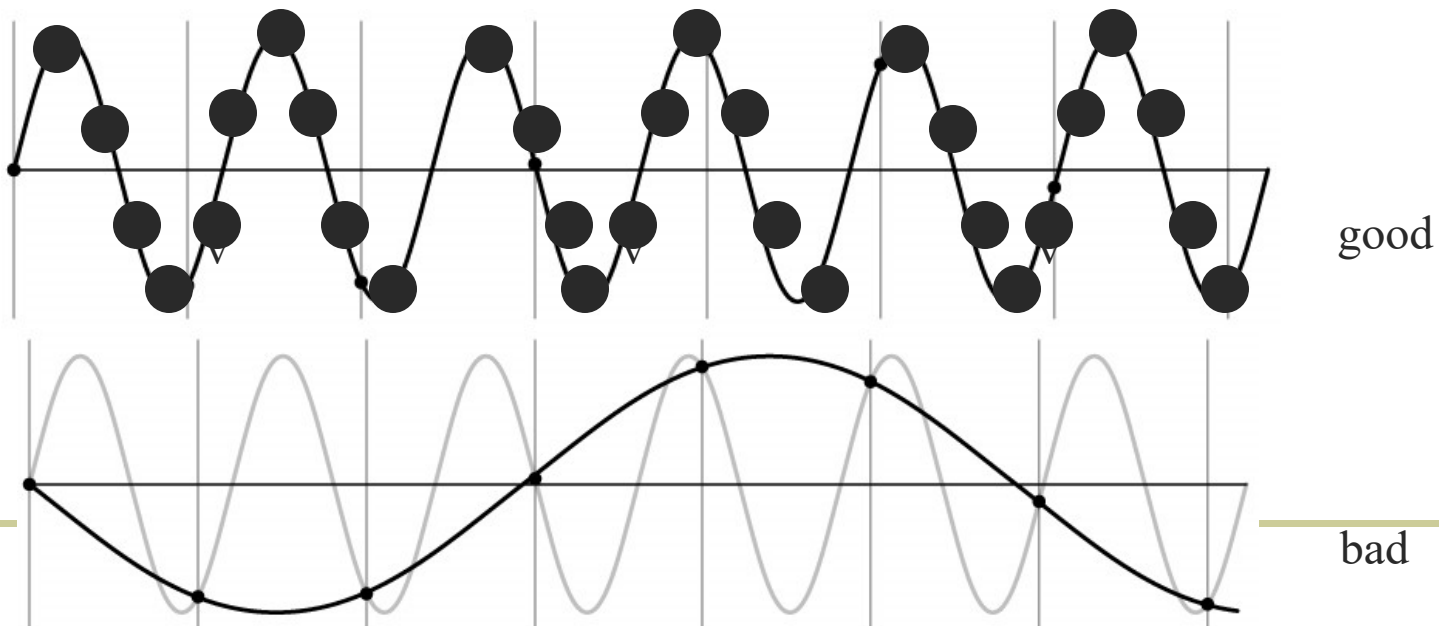




Nyquist-Shannon Sampling Theorem



- When sampling a signal at discrete intervals, the sampling frequency must be $\geq 2 \times f_{\max}$
- $f_{\max}$ = max frequency of the input signal
- This will allow to reconstruct the original perfectly from the sampled version





Anti-aliasing



Solutions:

- Sample more often
- Get rid of all frequencies that are greater than half the new sampling frequency
 - Will lose information
 - But it's better than aliasing
 - Apply a smoothing filter



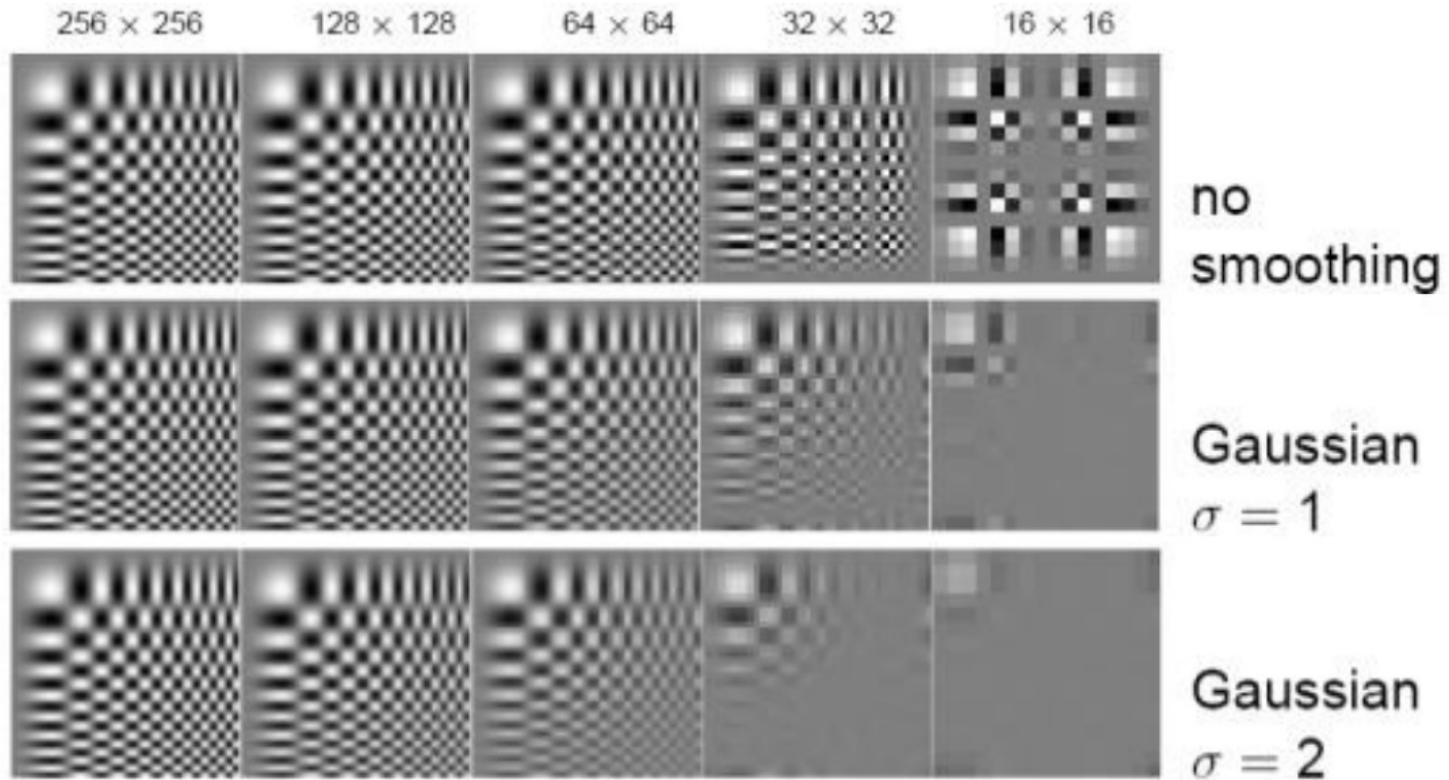
Algorithm for downsampling by factor of 2



1. Start with image(h, w)
2. Apply low-pass filter
`im_blur = imfilter(image, fspecial('gaussian', 7, 1))`
3. Sample every other pixel
`im_small = im_blur(1:2:end, 1:2:end);`



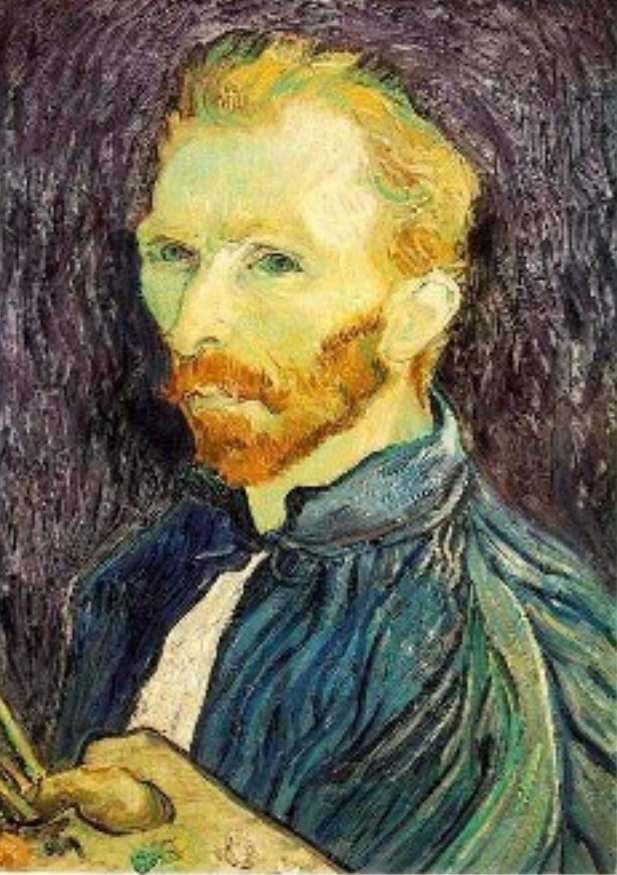
Anti-aliasing



Note: We cannot recover the high frequencies, but we can avoid artifacts by smoothing before resampling.



Subsampling without pre-filtering



$1/2$



$1/4$ (2x zoom)



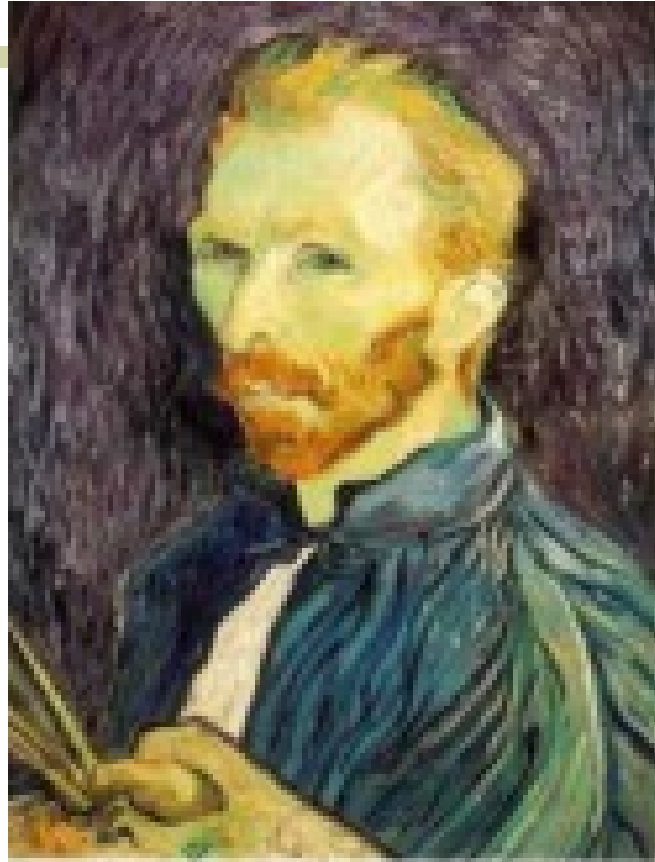
$1/8$ (4x zoom)



Subsampling with Gaussian pre-filtering



Gaussian $1/2$



G $1/4$



G $1/8$



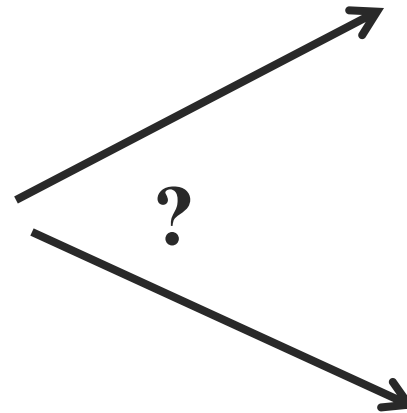
Today's Class



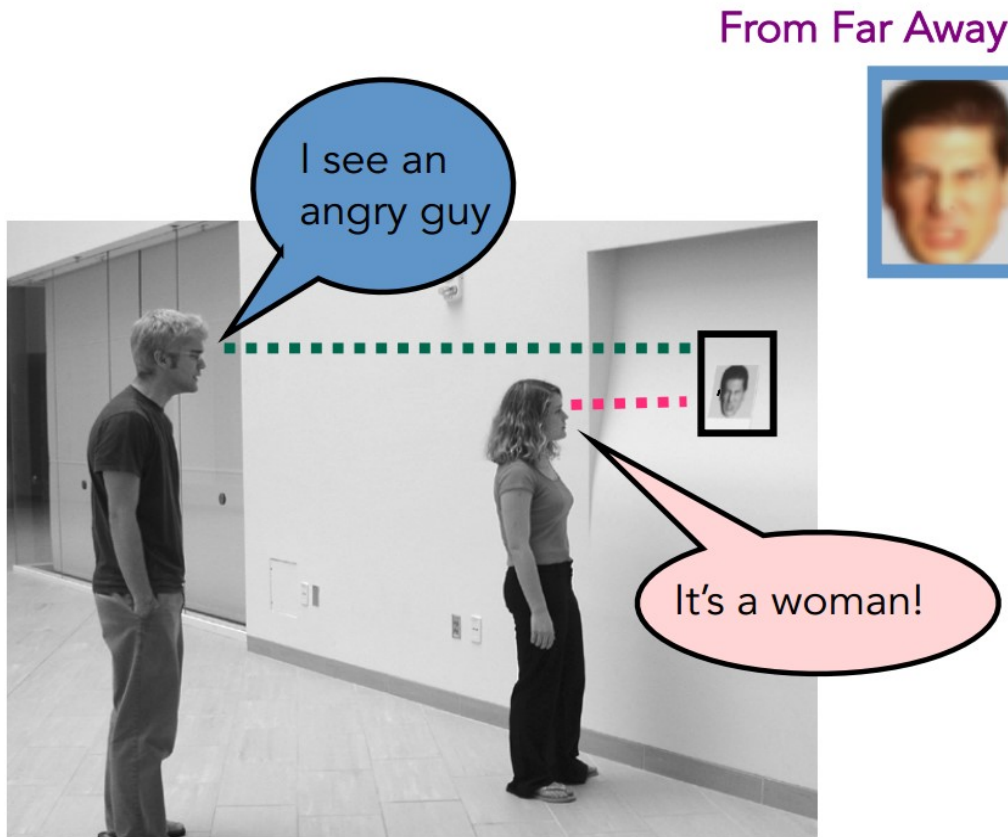
- Frequency domain and Fourier transform
- 1D Discrete Fourier Transform
- 2D Image Fourier Transform
- Sampling
- **Hybrid Image**



Why do we get different, distance-dependent interpretations of hybrid images?



Hybrid Images

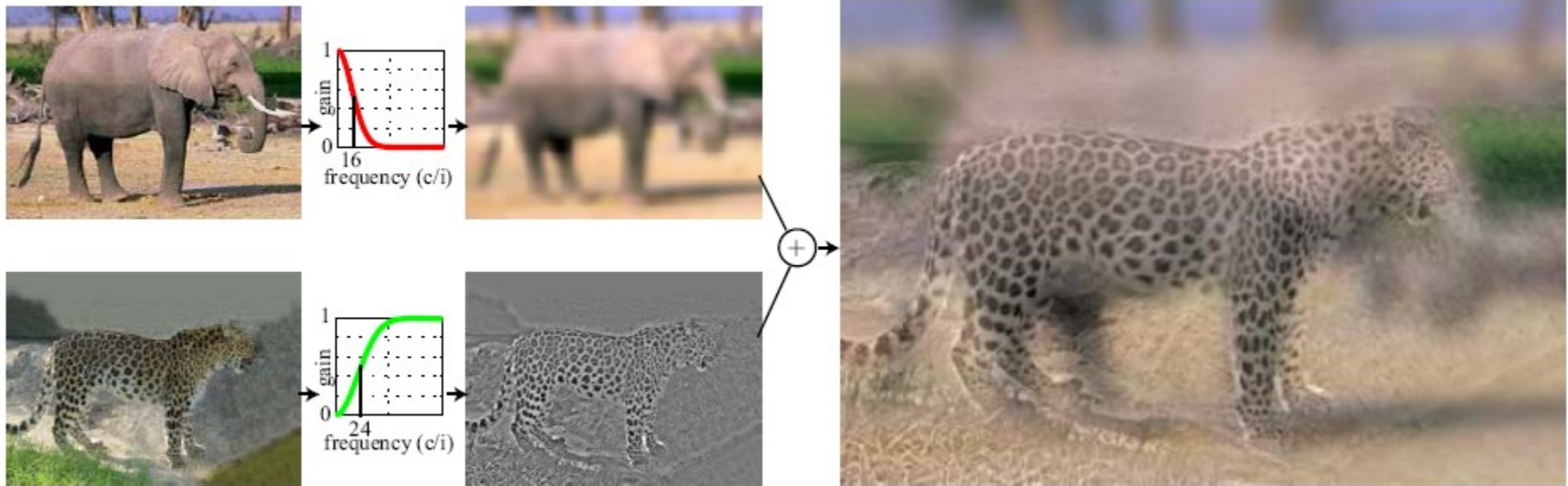


Up Close





Hybrid Images



- A. Oliva, A. Torralba, P.G. Schyns, “Hybrid Images,” SIGGRAPH 2006

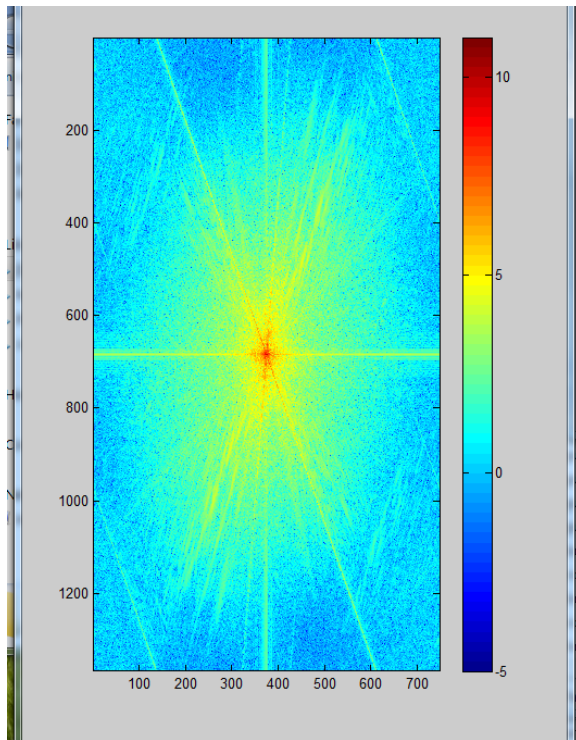




Hybrid Image in FFT

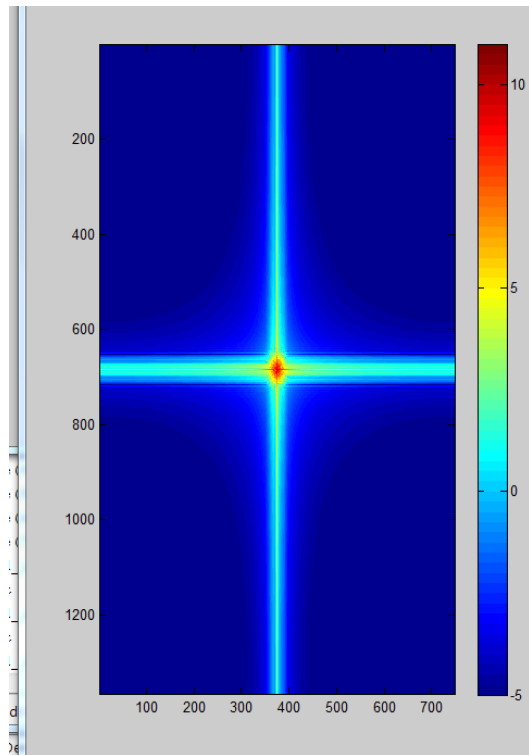


Hybrid Image

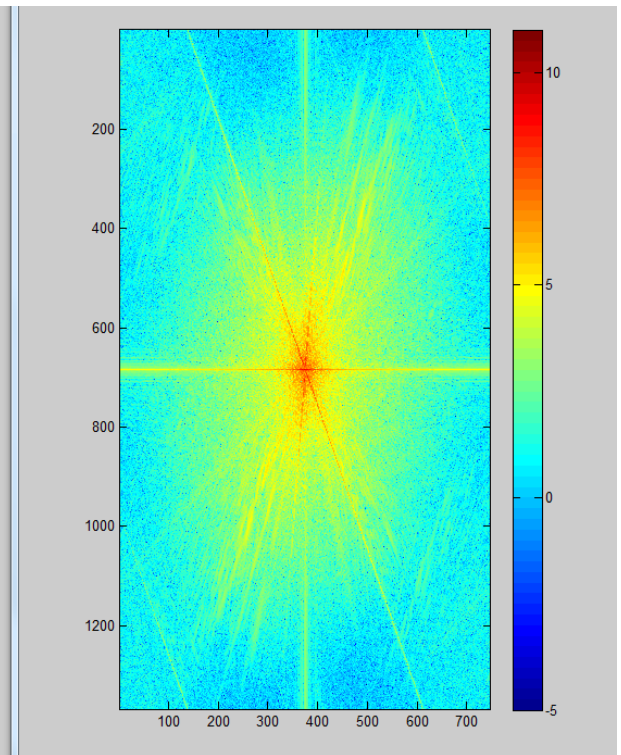


=

Low-passed Image



High-passed Image





Campbell & Robson chart



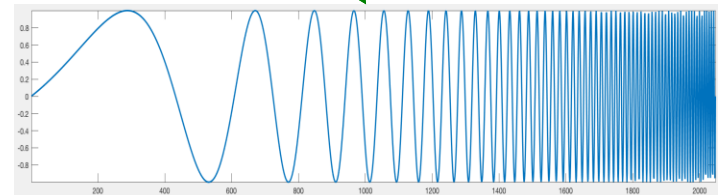
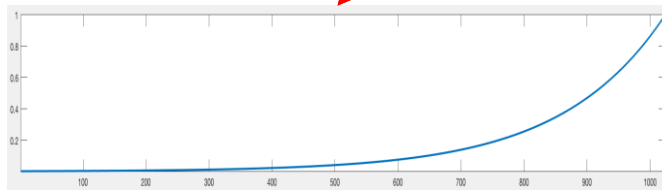
Let's define the following image:

$$I[n, m] = A[n] \sin(2\pi f[m] m/M)$$

With:

$$A[n] = A_{min} \left(\frac{A_{max}}{A_{min}} \right)^{n/N}$$

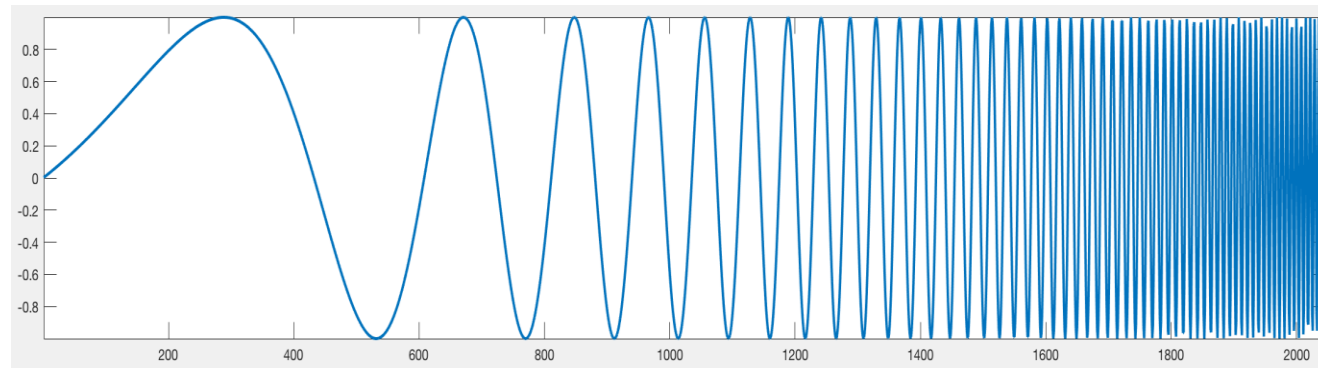
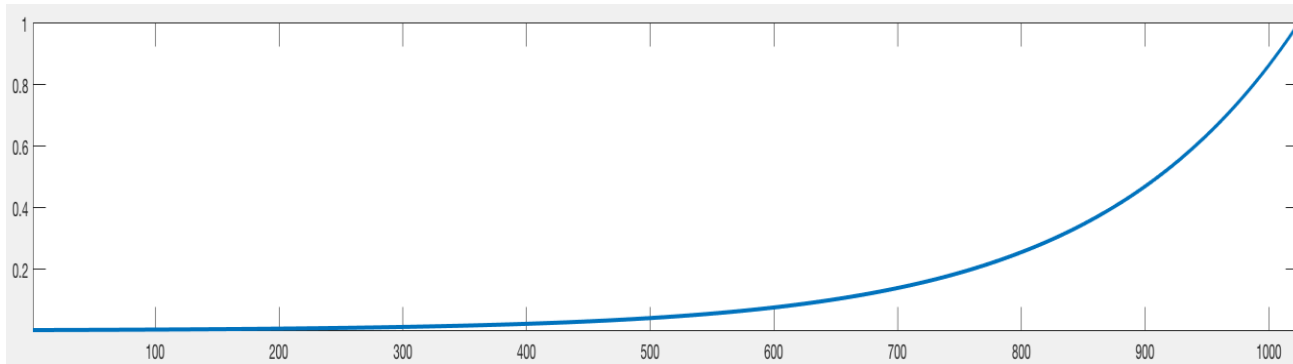
$$f[m] = f_{min} \left(\frac{f_{max}}{f_{min}} \right)^{m/M}$$



What do you think you should see when looking at this image?

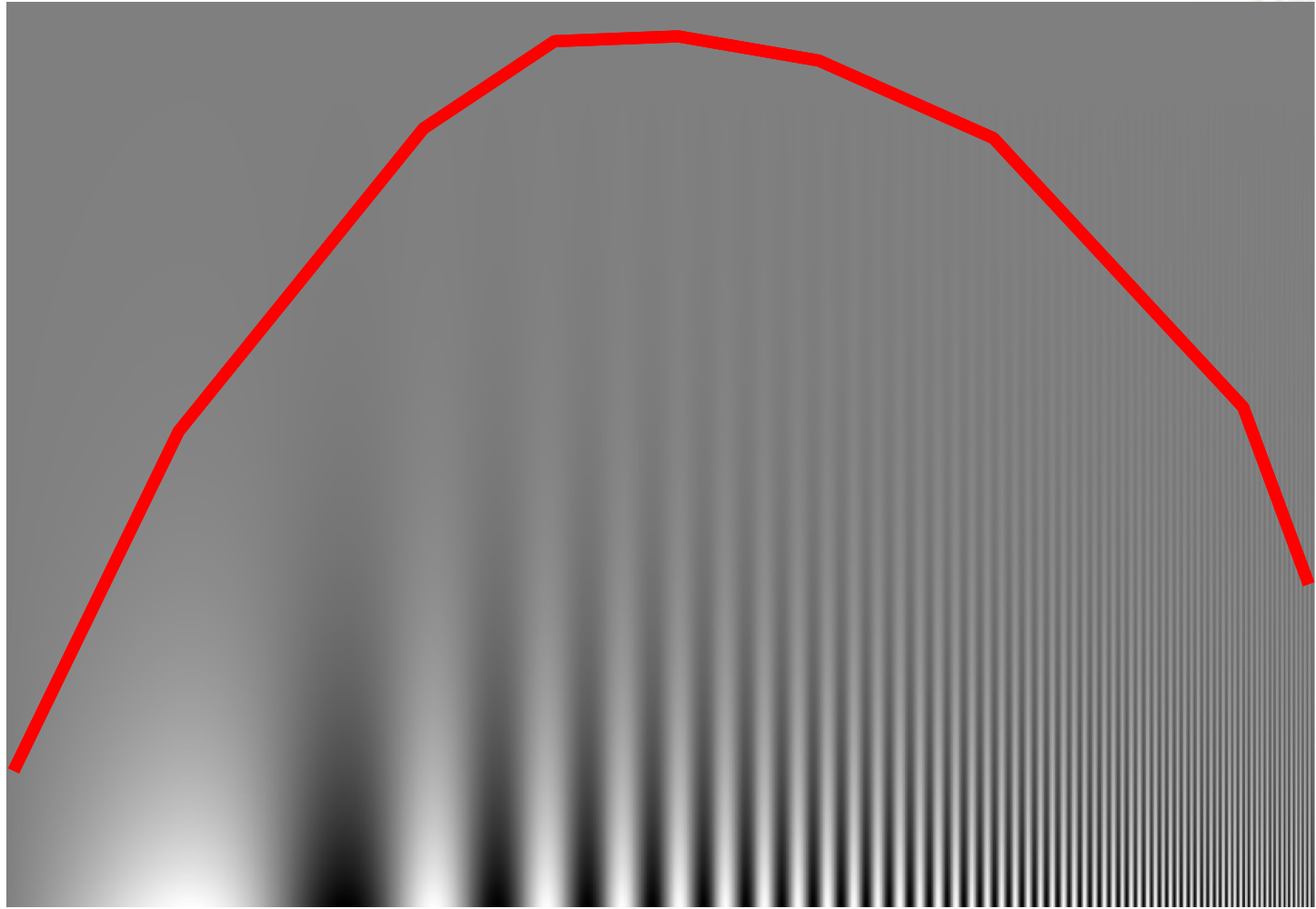


$$\mathbf{I}[n, m] = A[n] \sin(2\pi f[m] m/M)$$





$$\mathbf{I}[n, m] = A[n] \sin(2\pi f[m] m/M)$$





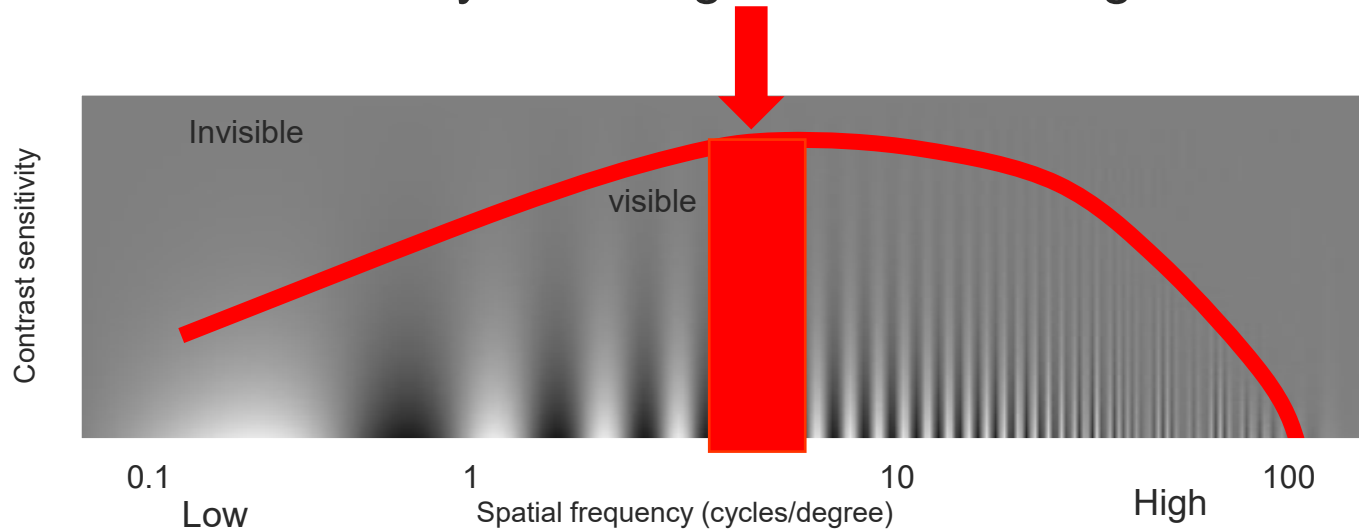
Contrast Sensitivity Function



Blackmore & Campbell (1969)

Maximum sensitivity

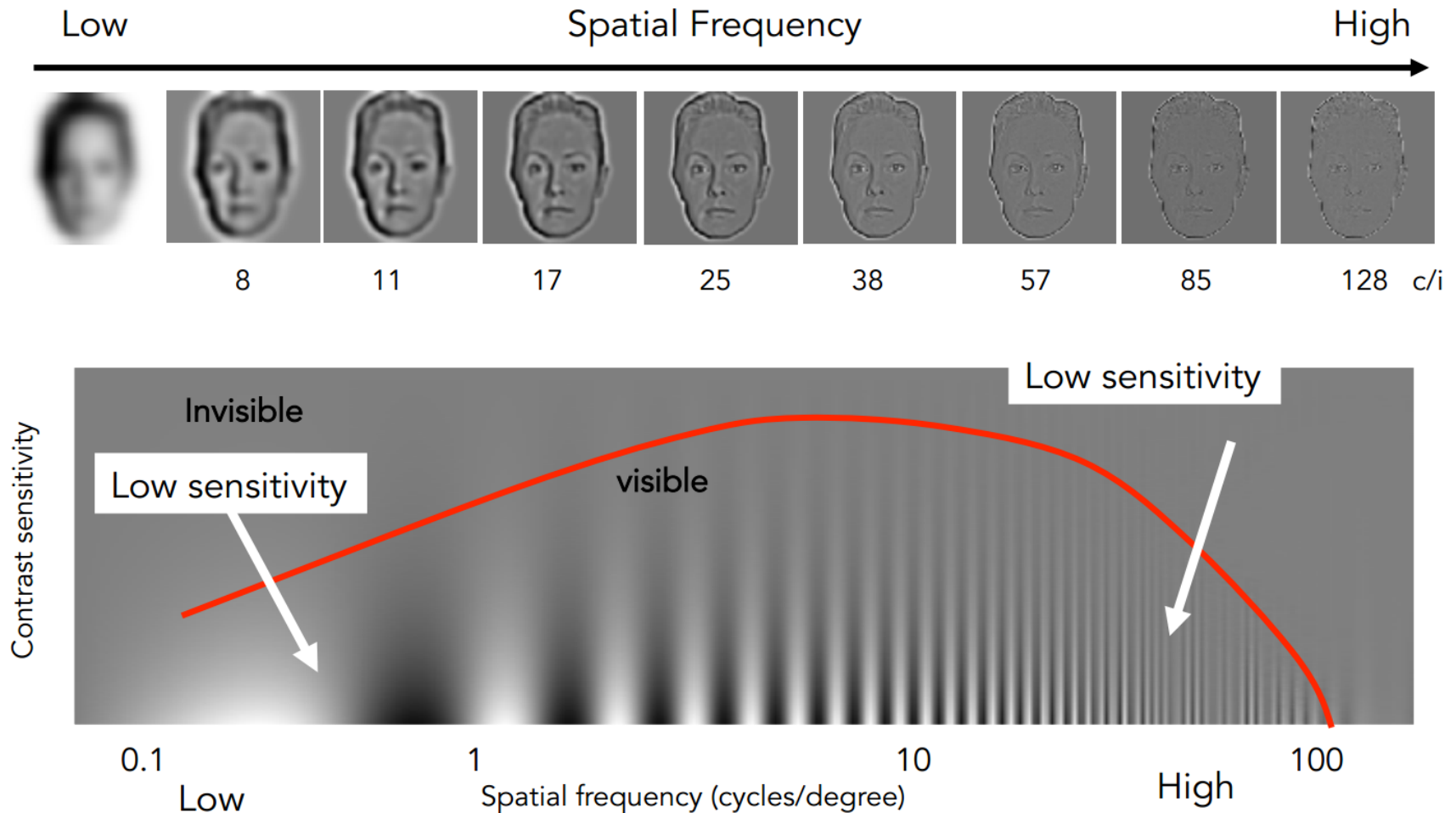
~ **6** cycles / degree of visual angle



Things that are very close
and/or large are hard to see

Things far away
are hard to see

Multiscale subband decomposition + Human Contrast Sensitivity Function



Contrast Sensitivity Function

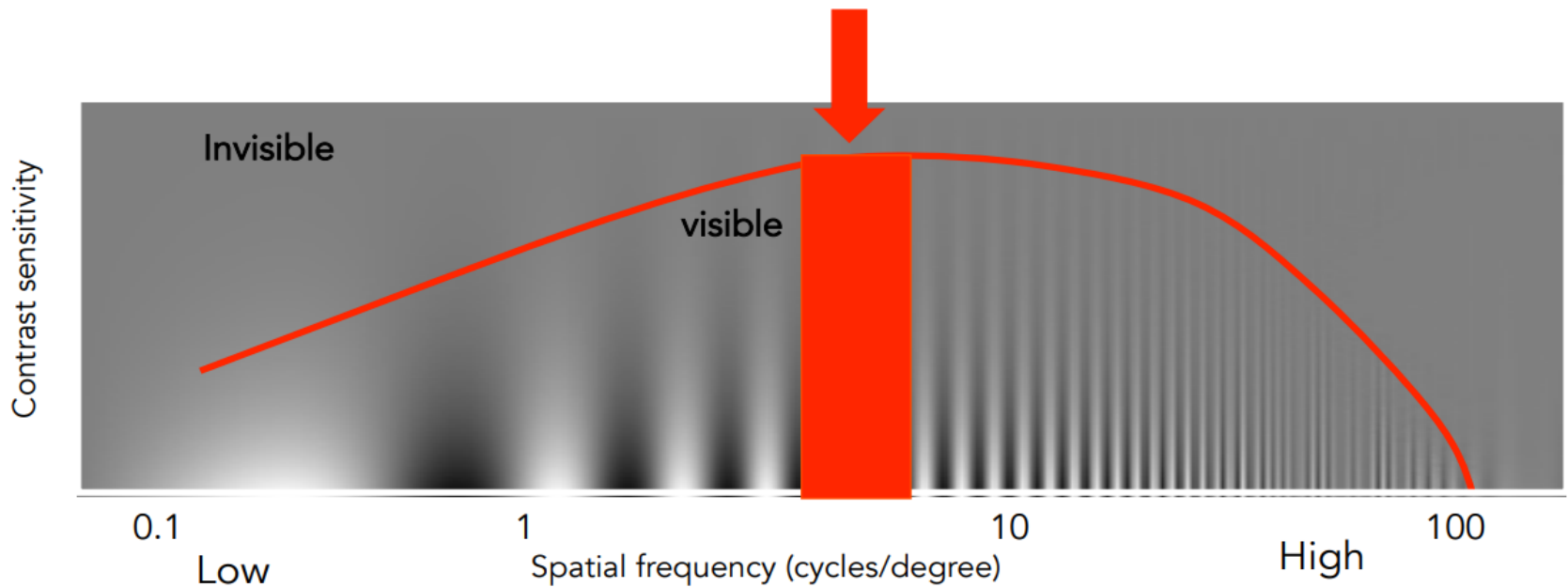
Low

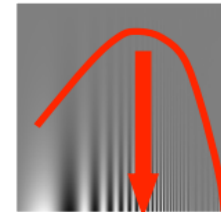
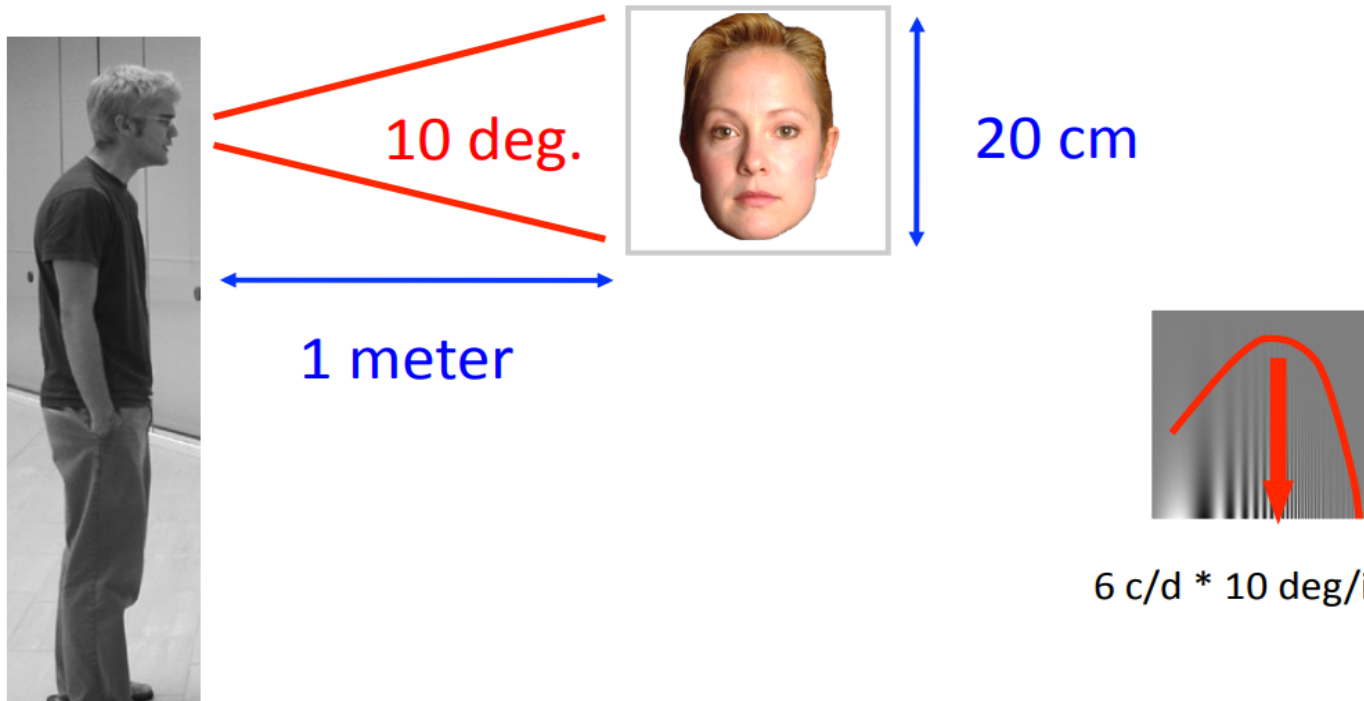


High

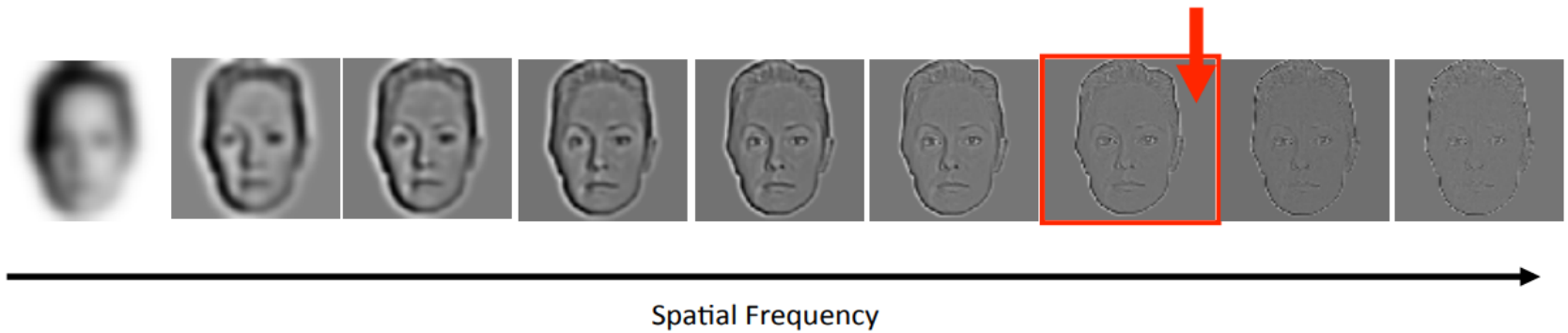


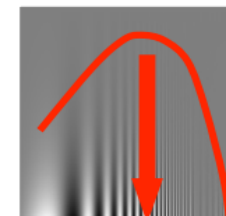
Maximum sensitivity
~ 6 cycles / degree of visual angle



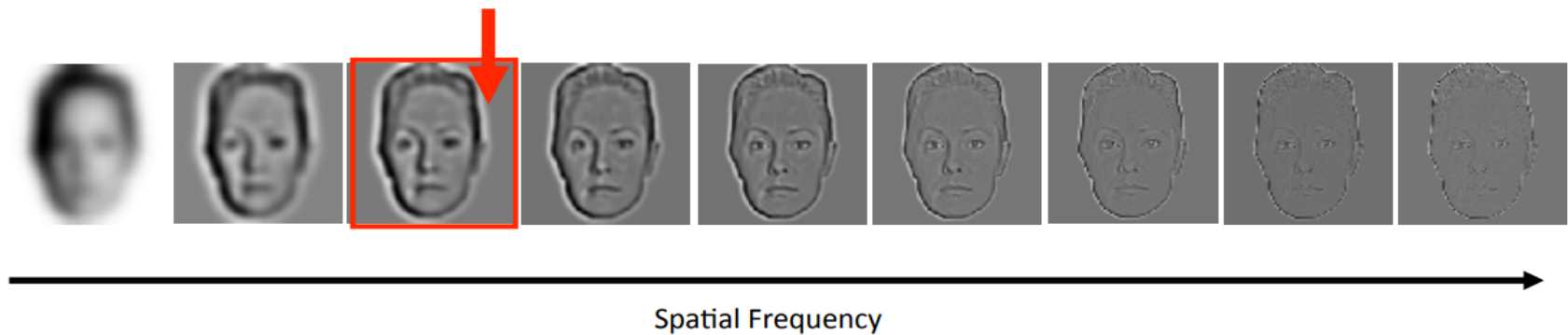


$$6 \text{ c/d} * 10 \text{ deg/i}$$





$$6 \text{ c/d} * 2 \text{ deg/i}$$

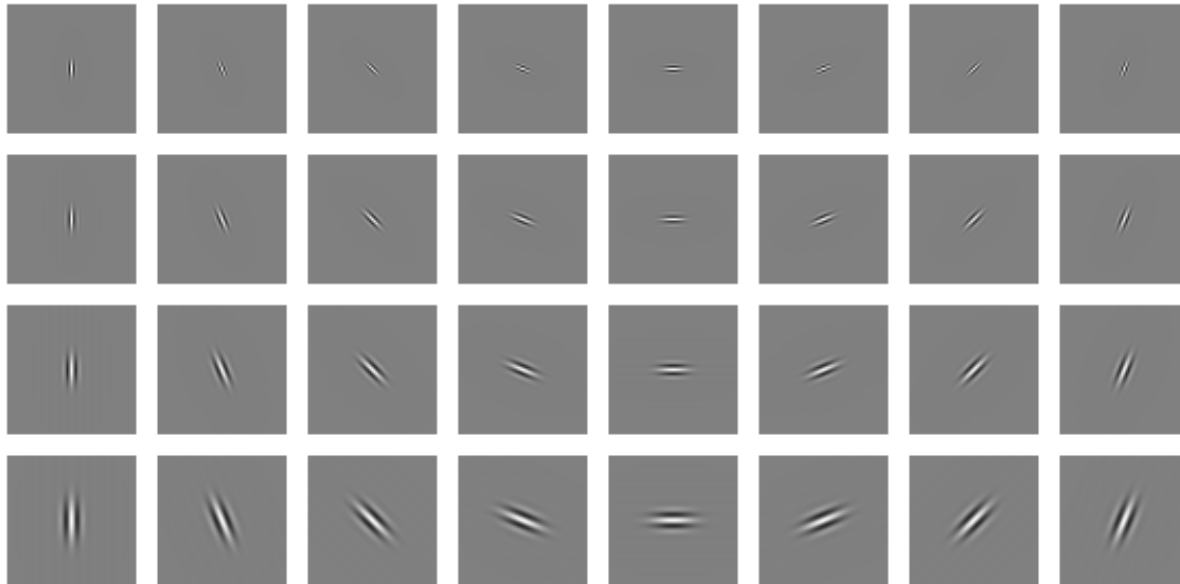




Clues from Human Perception



- Early processing in humans filters for various orientations and scales of frequency
- Perceptual cues in the mid-high frequencies dominate perception



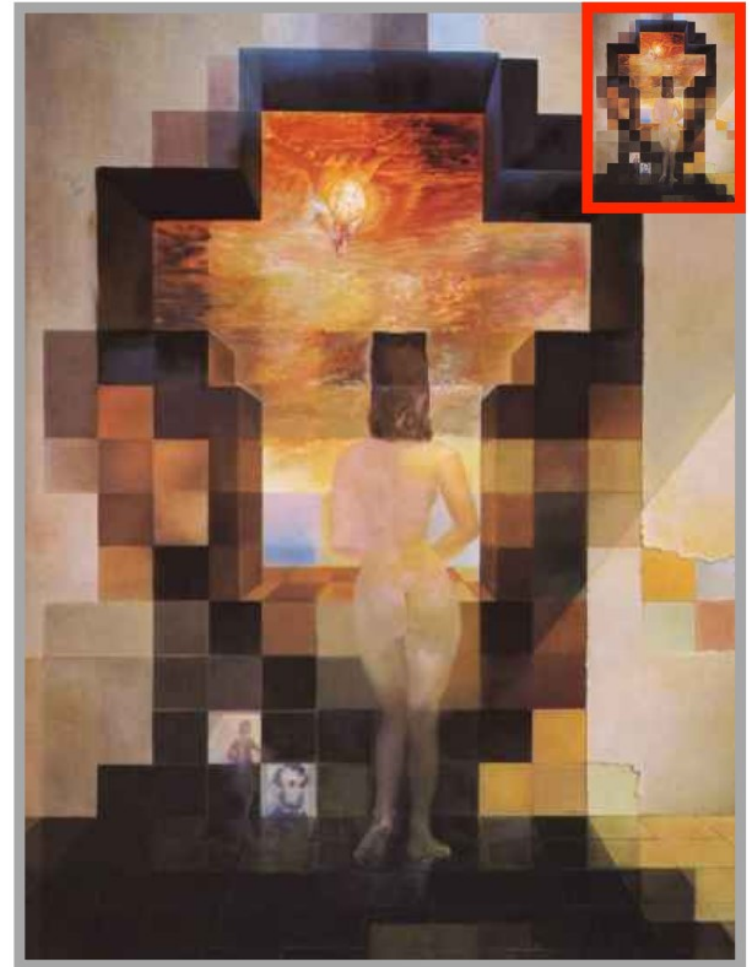
Early Visual Processing: Multi-scale edge and blob filters

Visual Perception

For computer vision systems to “understand art” other properties of human visual perception must be taken into account



Salvador Dali



Changes of Meaning with Scale

Strong percept of a skull



Strong percept of a woman



Looking at the image one way, a skull appears, comprised of the woman's form and her reflection in a mirror (the outline of the skull itself).



"All Is Vanity", by C. Allan Gilbert 1873-1929.



Things to Remember

- Sometimes it makes sense to think of images and filtering in the frequency domain
 - Fourier analysis
- Can be faster to filter using FFT for large images ($N \log N$ vs. N^2 for auto-correlation)
- Remember to low-pass before sampling

